

LSST DSFP, June 2026

Shift Happens: Building Robust AI Models with Domain Adaptation

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U.S. DEPARTMENT
of **ENERGY**

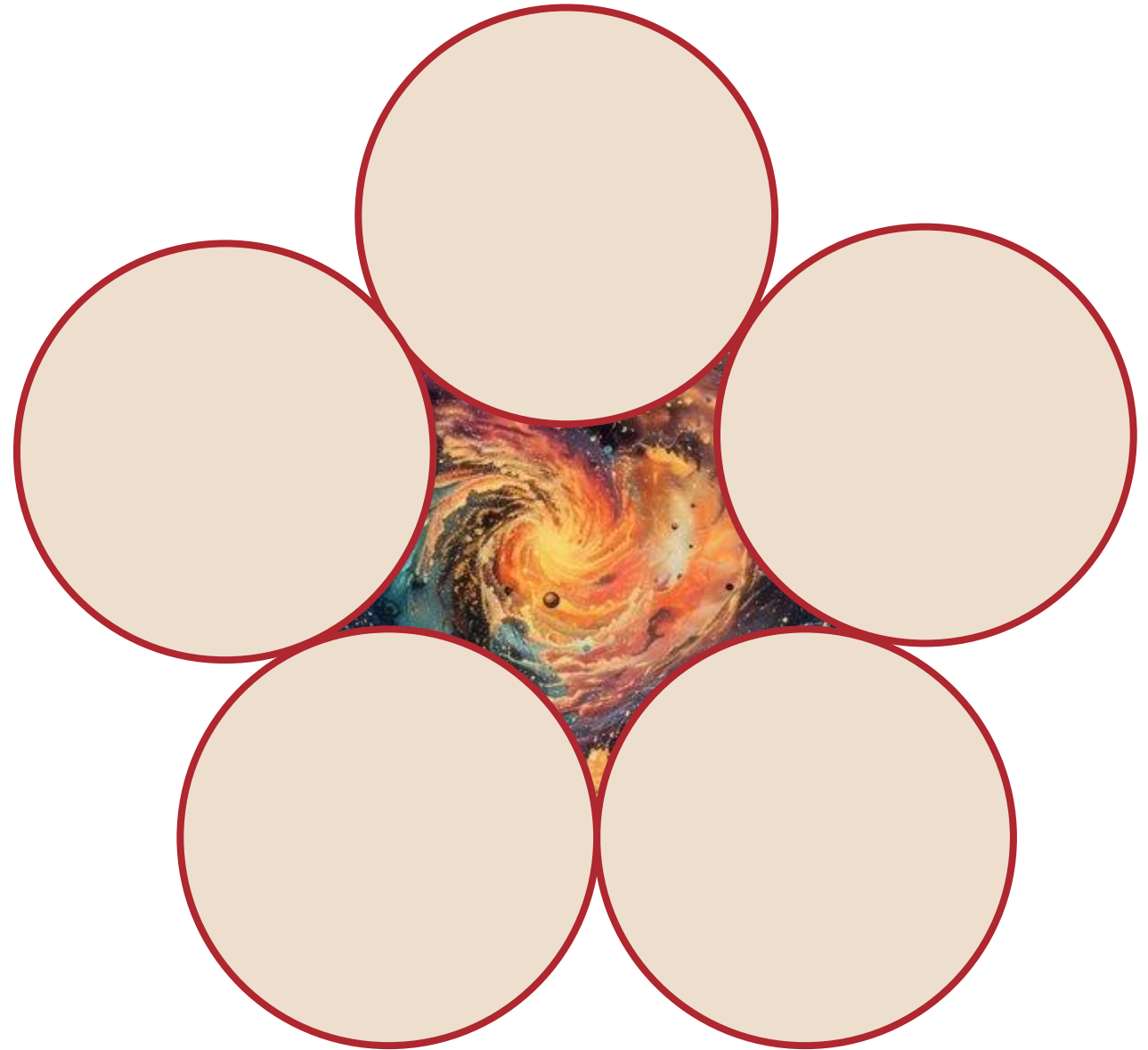
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Major challenges for AI applications in Science

As listed in The Dawes Review 10: The impact of deep learning for the analysis of galaxy surveys
[Huertas-Company & Lanusse 2022](#)





Major challenges for AI applications in Science

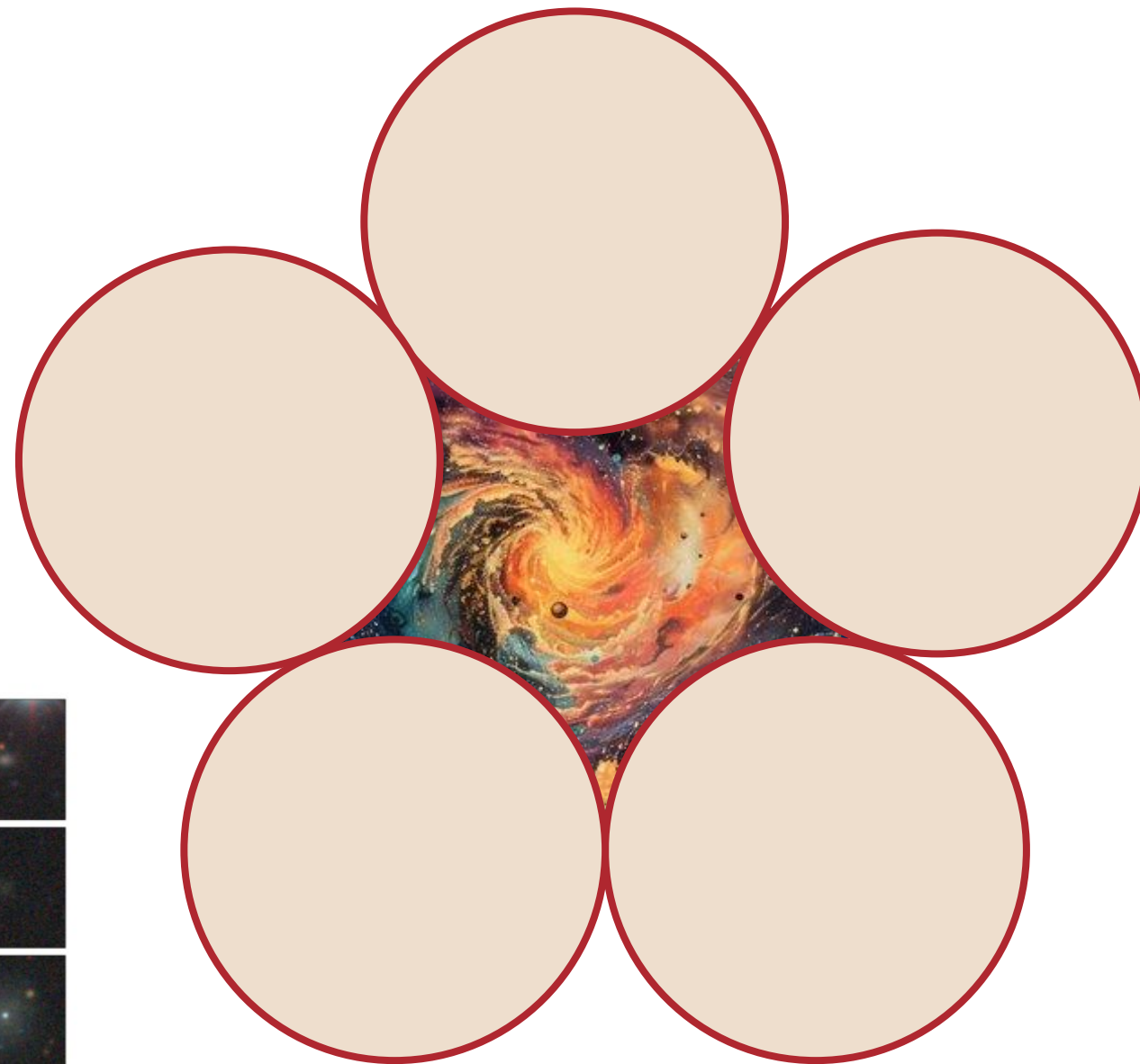
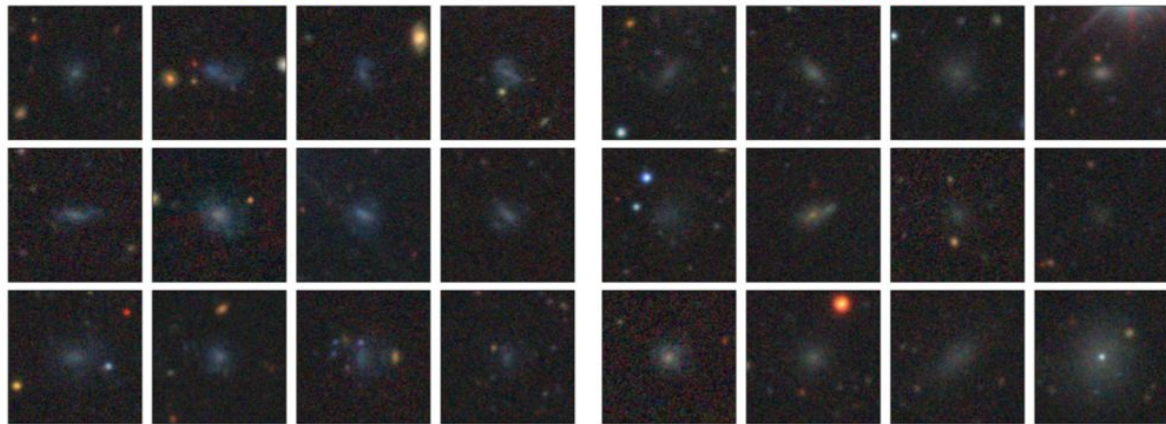
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Low Surface Brightness Galaxies (LSBGs)

Tanogldis et al. 2021.

Blue LSBGs

Red LSBGs





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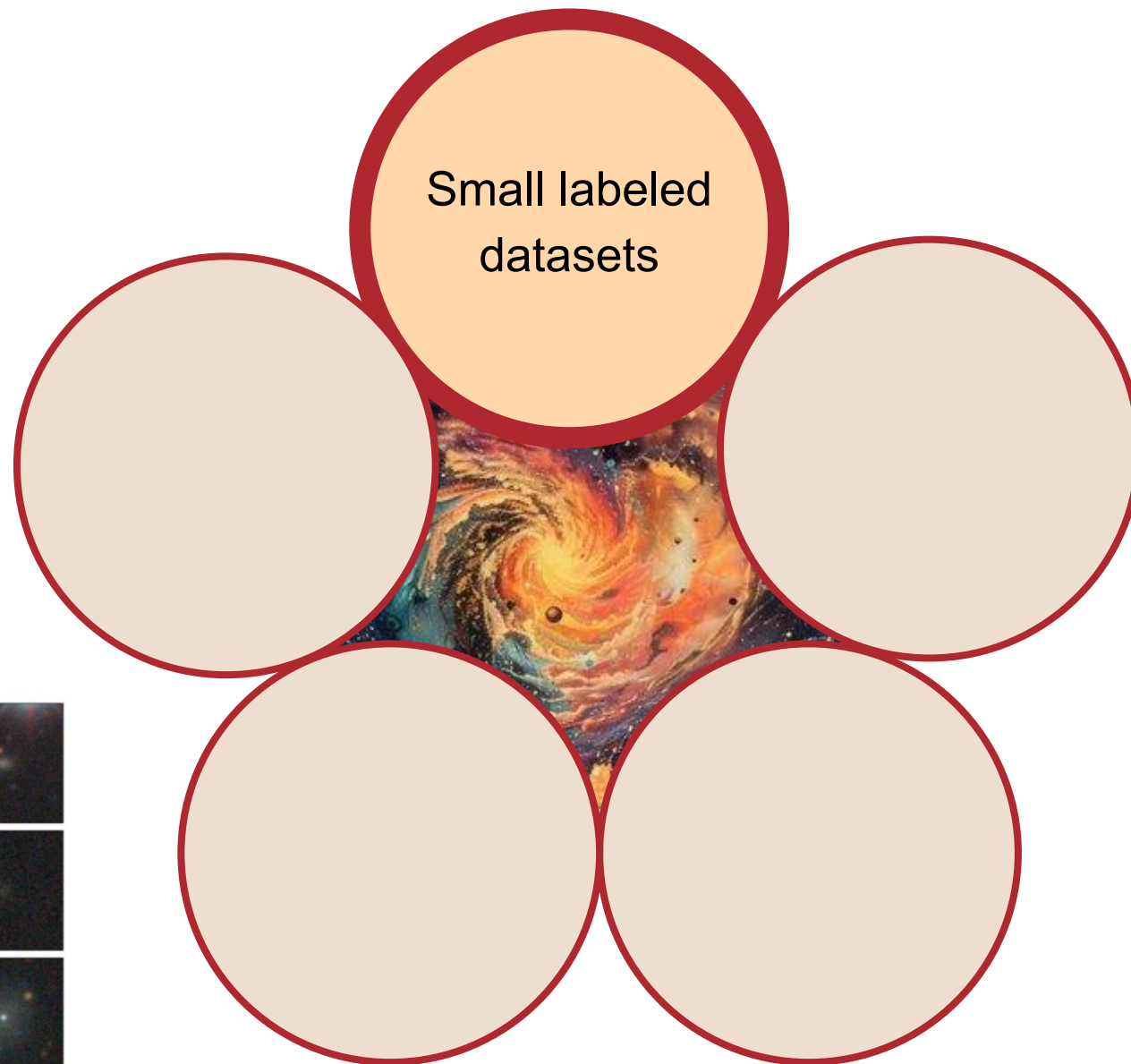
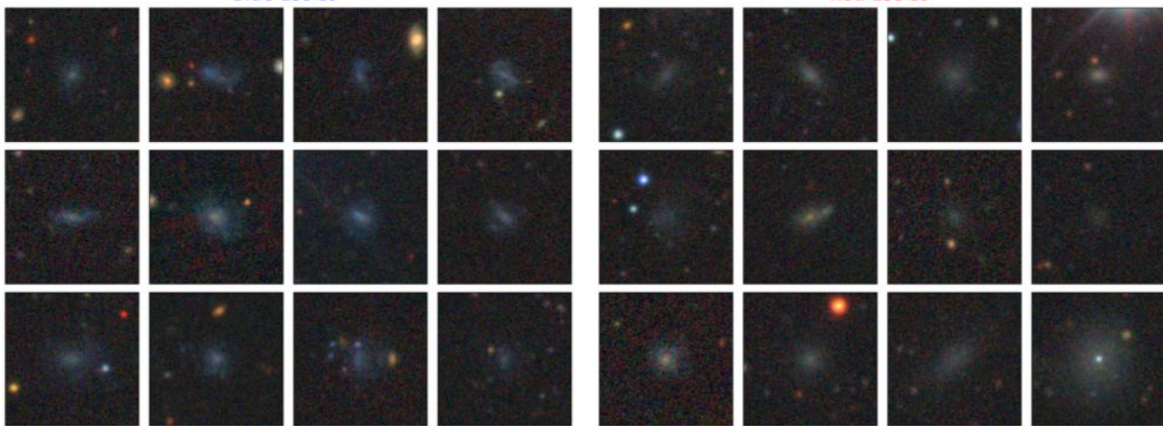
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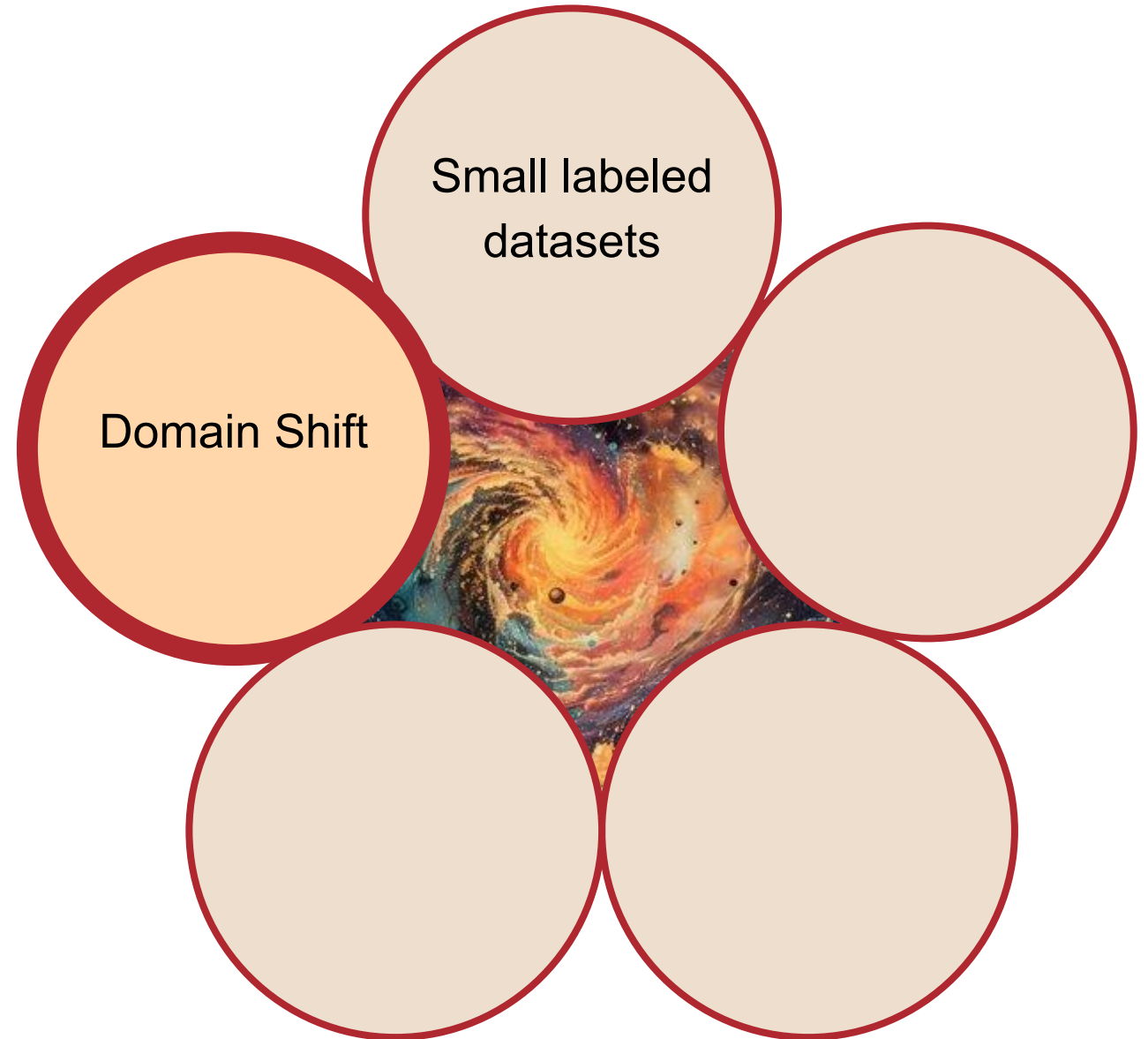


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Can I train on simulations and apply the model to my real data?





Major challenges for AI applications in Science

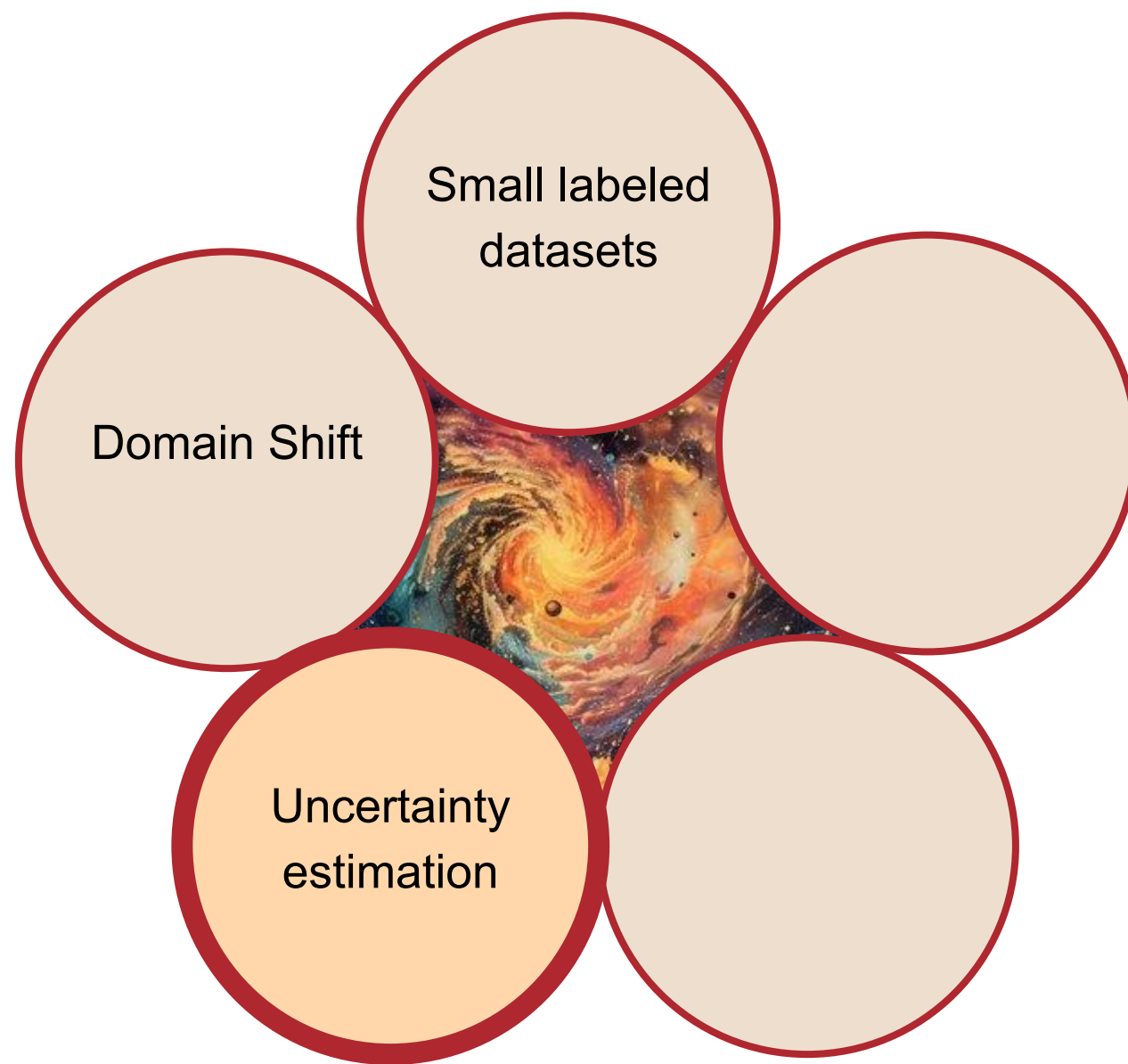
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How well is my model going to perform?





Major challenges for AI applications in Science

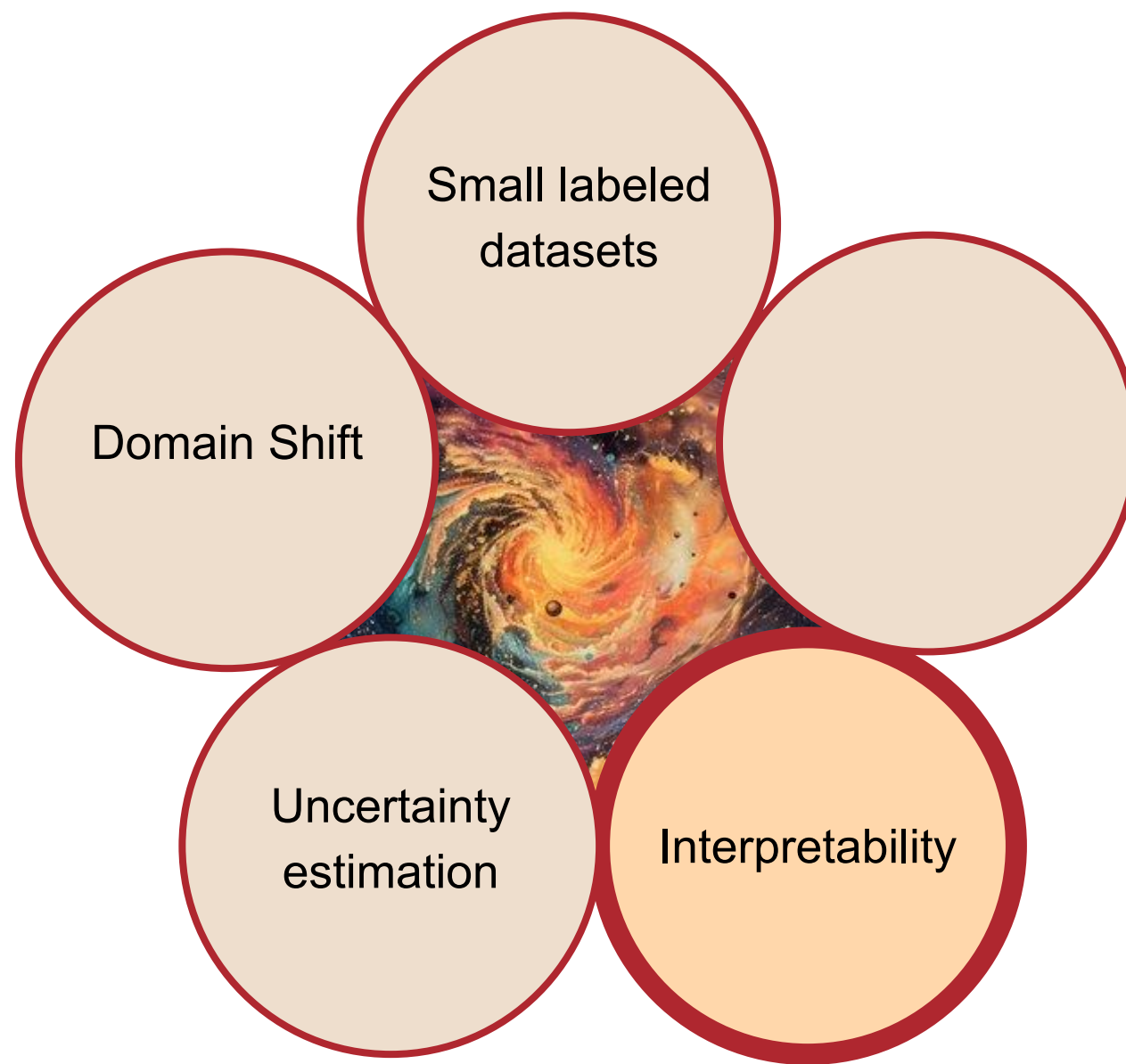
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How well is my model going to perform?

Can I understand its performance?





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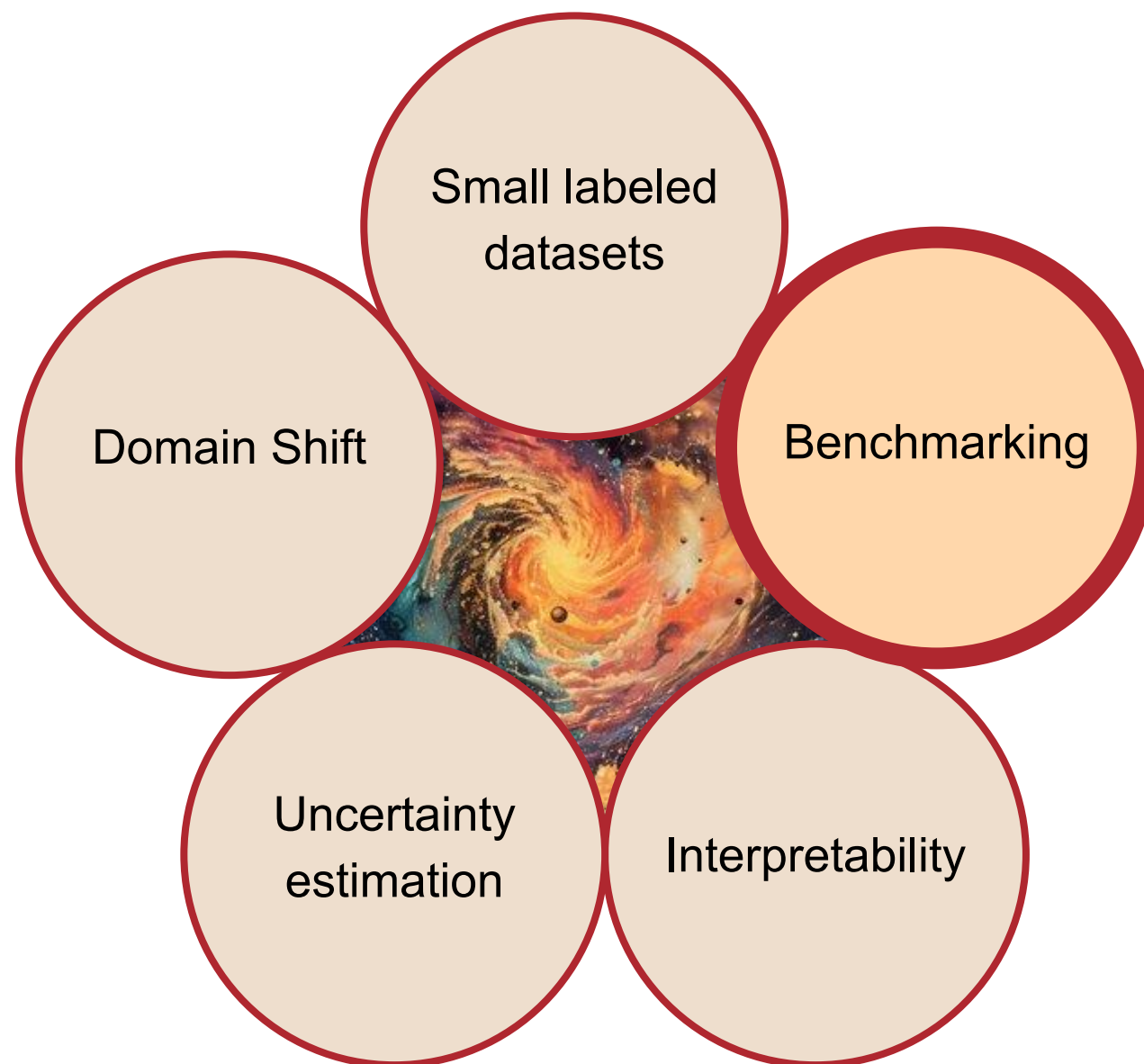
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Can I compare my model with other researchers?





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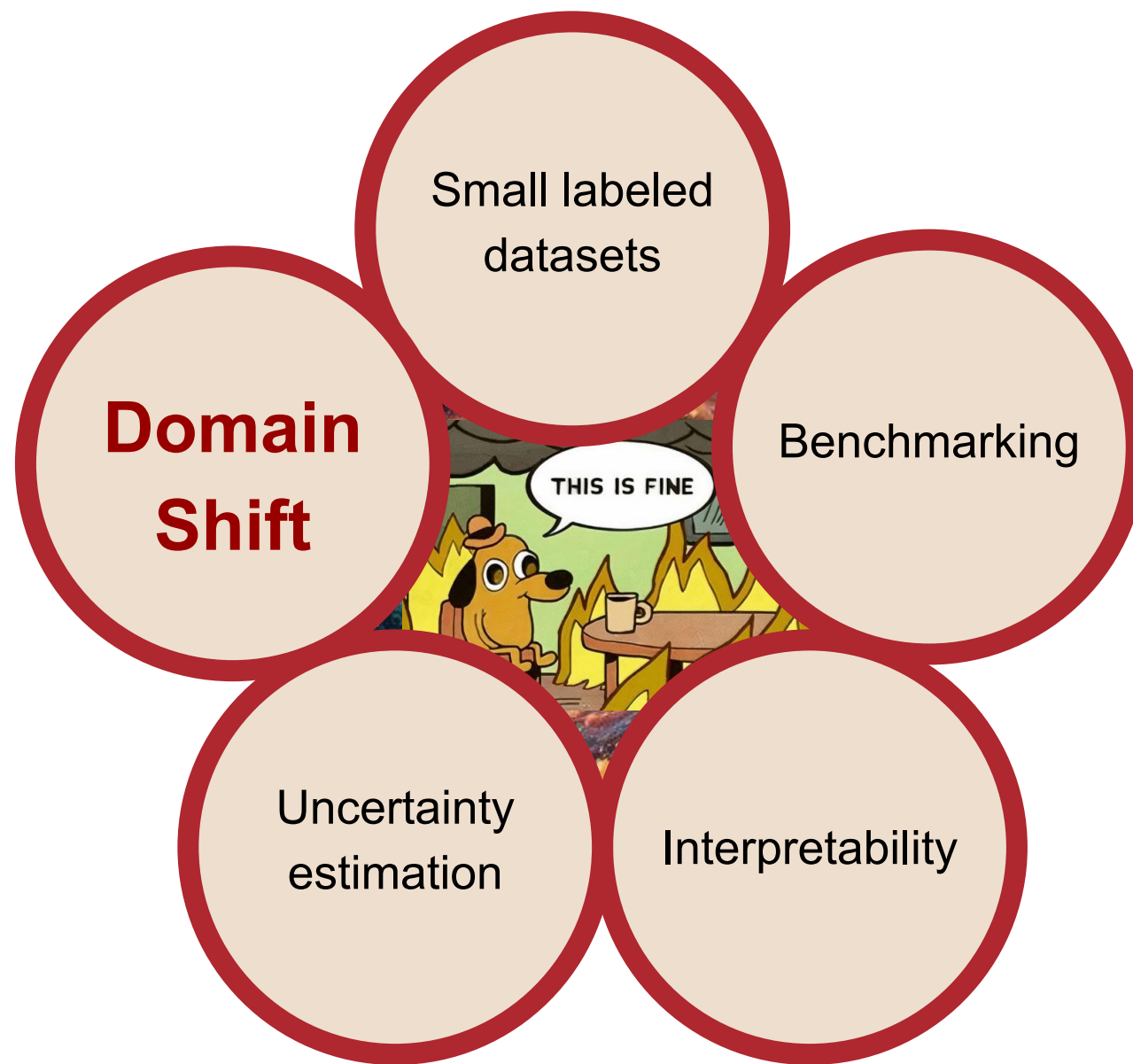
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Talk Outline

Part 1: Domain Shift Problem

Part 2: Domain Adaptation Techniques

Part 3: Examples

Part 4: Open Questions & Research Trends



01

Domain Shift Problem



More Data - More problems

All areas of astro(physics) often need to create **model trained on simulated data, that also work on real data!**

DOMAIN SHIFT

Missing and unknown physics, wrong geometry, background levels

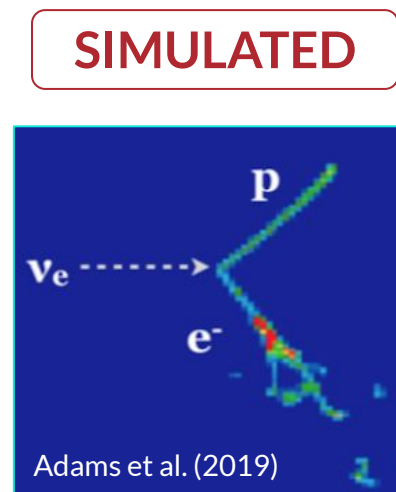
Computational constraints for simulations

Detector problems, transients, errors, data compression

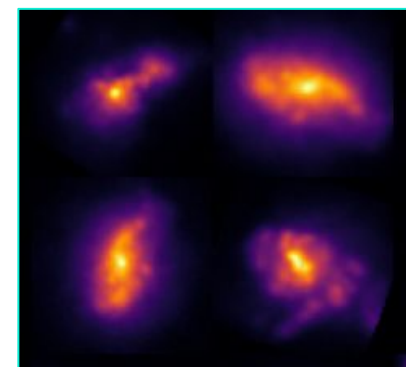
Imperfect addition of observational effects

Different detectors or telescopes

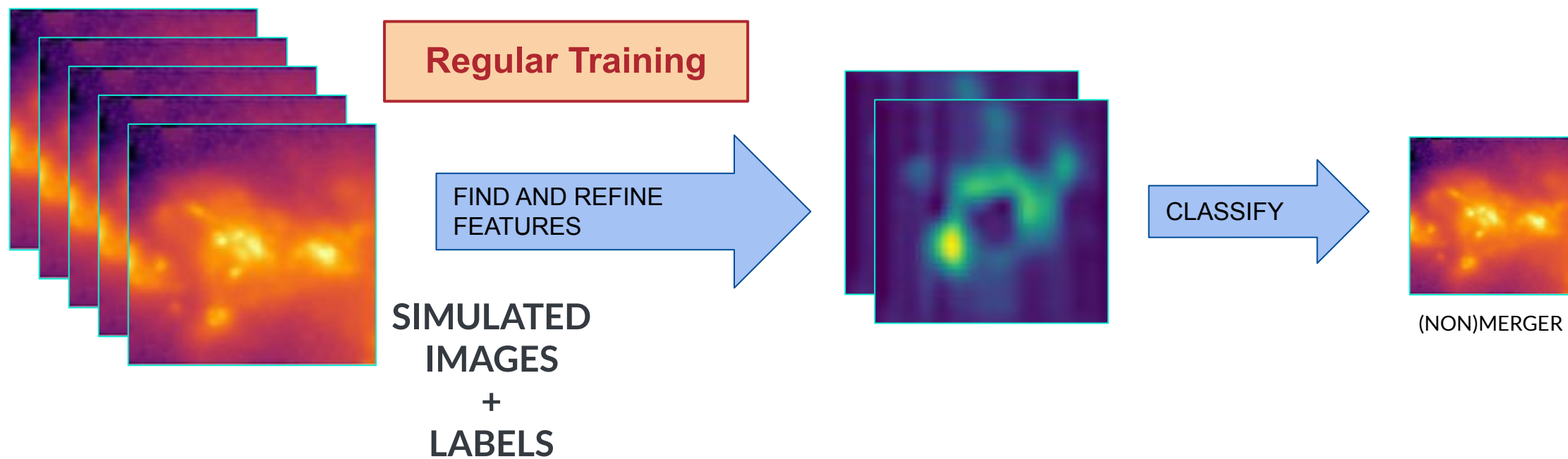
MicroBooNE
(neutrinos)



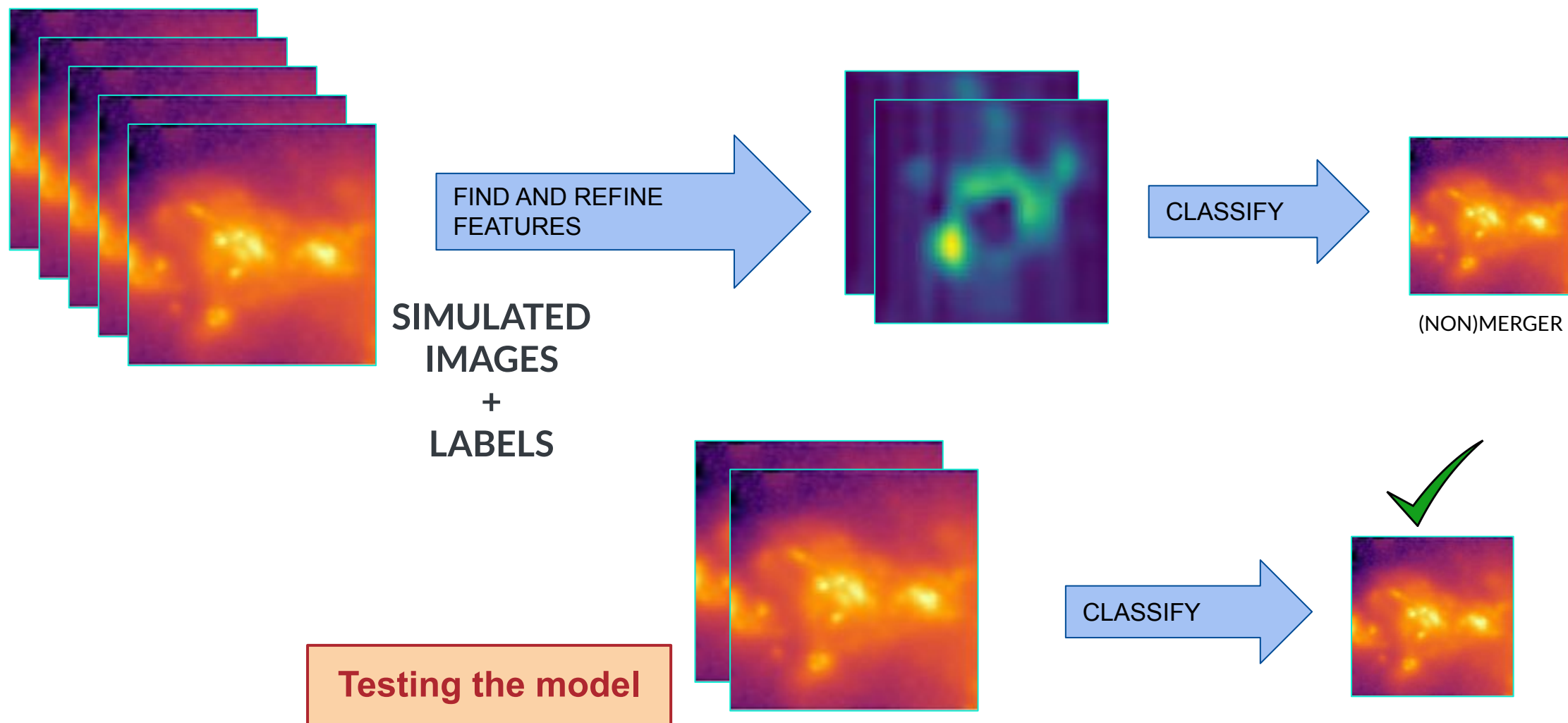
Illustris / Hubble
(merging galaxies)



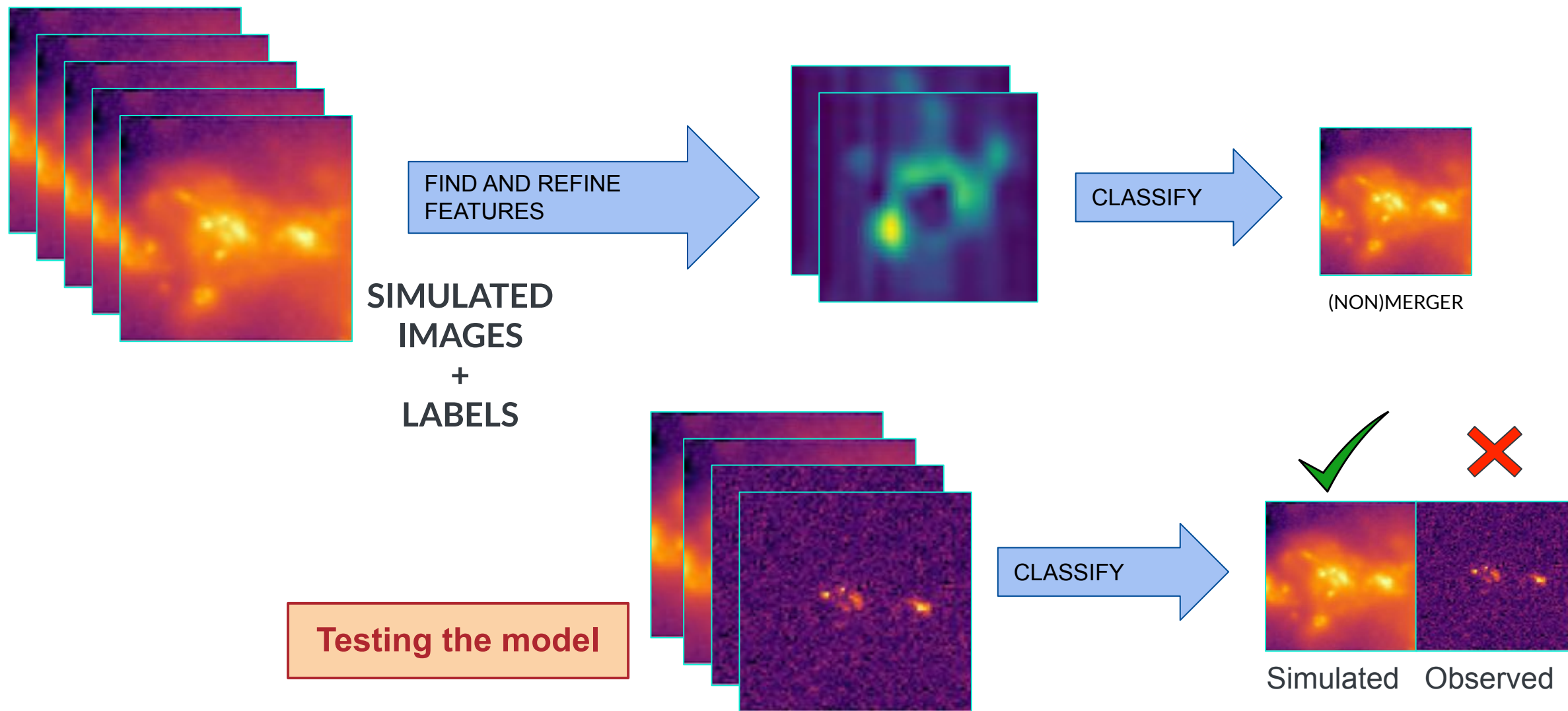
Traditional ML



Traditional ML



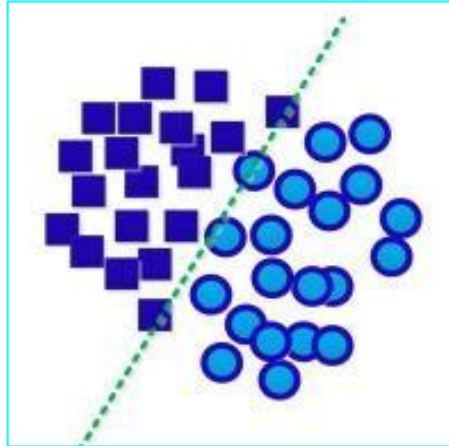
Traditional ML



Why do models fail?

Train the model
on source
dataset and find
the decision
boundary.

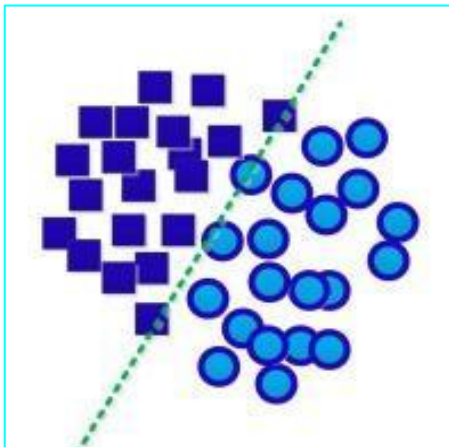
Source Domain



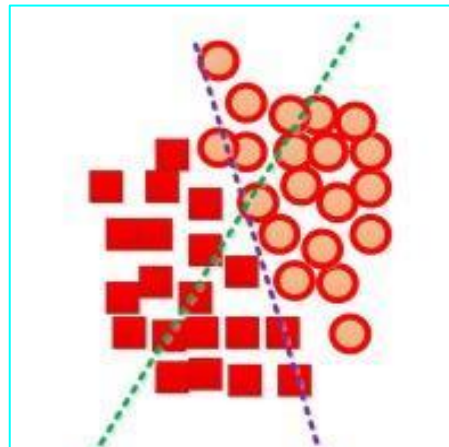
Why do models fail?

New domain is shifted,
learned decision
boundary doesn't
work.

Source Domain



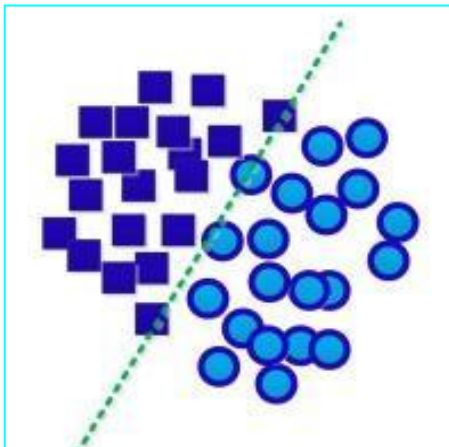
Target Domain



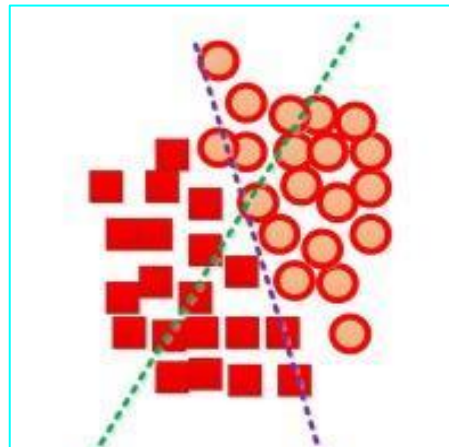
⚙️ Why do models fail?

We need to align
the data during
training!

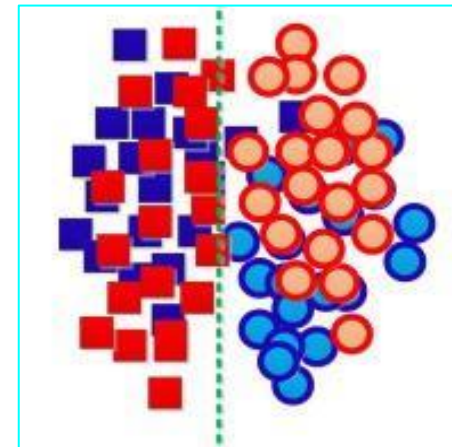
Source Domain



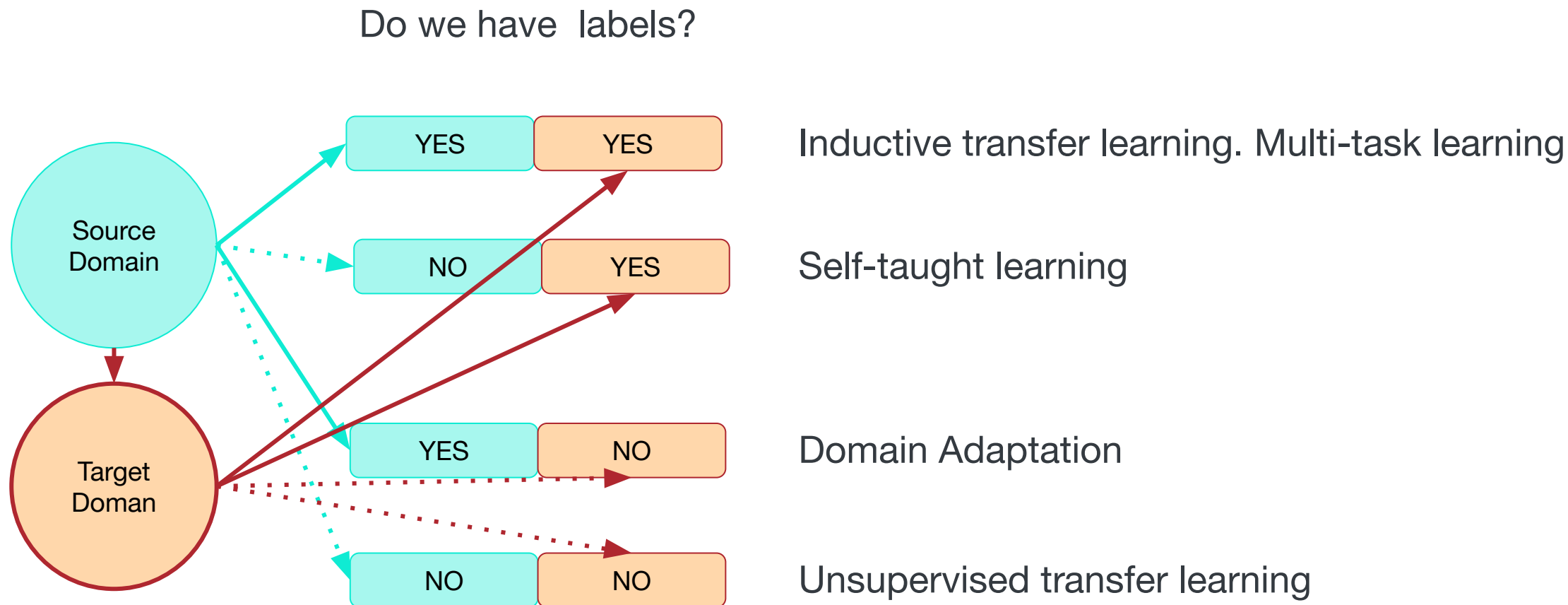
Target Domain



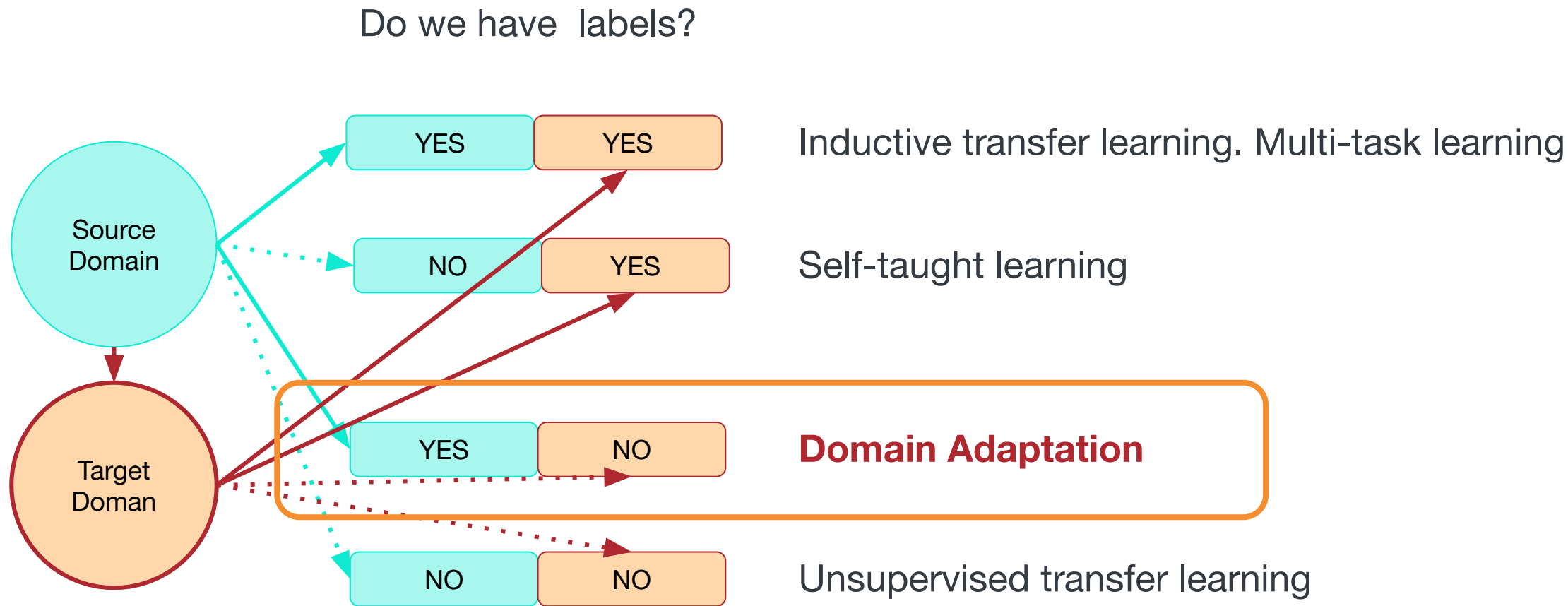
Domain Alignment



The Landscape of Multi-Dataset Learning

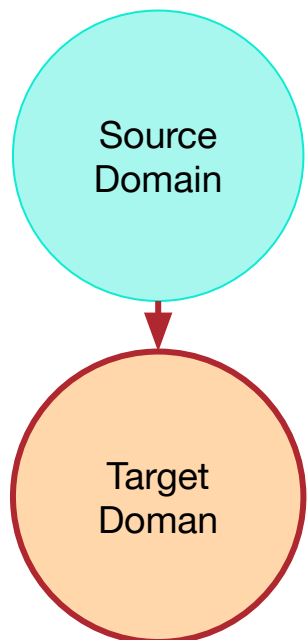


The Landscape of Multi-Dataset Learning





Datasets and Types of Shifts



$$\mathcal{D}_S = \{(x_i^S, y_i^S)\}_{i=1}^{n_S}$$

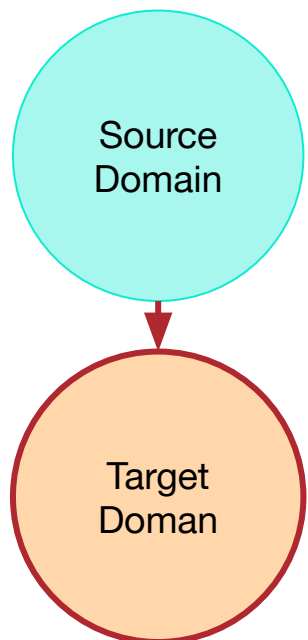
with n_S labelled samples, from joint distribution $P_S(x, y)$

$$\mathcal{D}_T = \{x_i^T\}_{i=1}^{n_T}$$

with n_T unlabelled samples, from distribution $P_T(x)$



Datasets and Types of Shifts



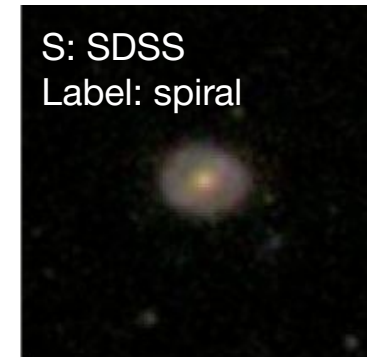
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S: SDSS
Label: spiral



T: SDSS
No Label



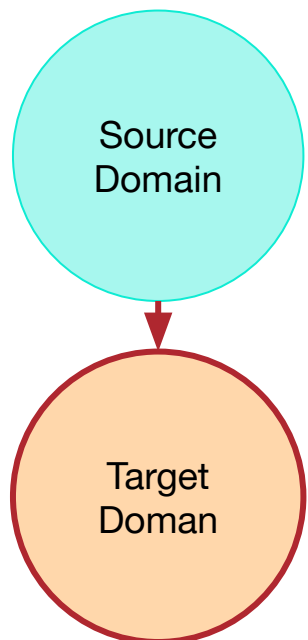
(we observe more
faint galaxies)

Label Shift:

$$P_s(Y) \neq P_t(Y) \quad \text{while} \quad P_s(X | Y) = P_t(X | Y)$$

Conditional distribution of data given a label stays the same, but the label distribution **changes** i.e., the frequency of classes differs across domains (e.g., class imbalance), but **each class "looks" the same in both domains**.

⚙ Datasets and Types of Shifts



$$\mathcal{D}_S = \{(x_i^S, y_i^S)\}_{i=1}^{n_S}$$

with n_S labelled samples, from joint distribution $P_S(x, y)$

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S: SDSS
Label: spiral

T: SDSS
No Label

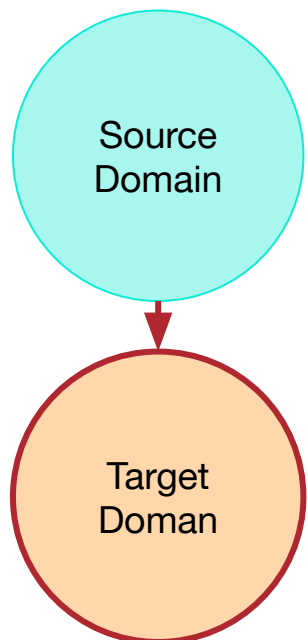
(barred/unbarred
spiral)

Concept Shift: $P_s(X) = P_t(X)$, but $P_s(Y | X) \neq P_t(Y | X)$

The input distribution of data stays the same, but the mapping from input to label **changes** i.e., the same input might be labeled differently across domains (e.g., due to different labeling policies or evolving concepts).



Datasets and Types of Shifts



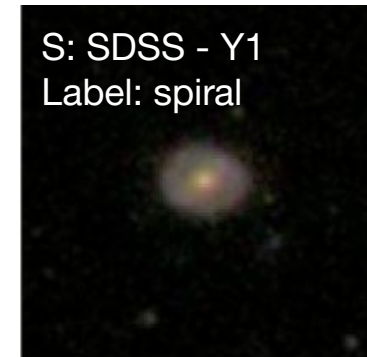
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S: SDSS - Y1
Label: spiral



T: SDSS - Y5
No Label

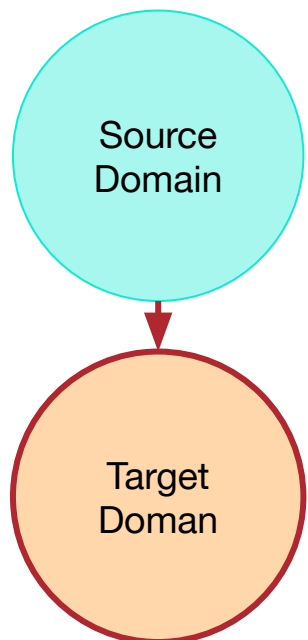


Conditional Shift:

$$P_S(Y) = P_T(Y) \quad \text{but} \quad P_S(X | Y) \neq P_T(X | Y)$$

The label distribution the same, but the mapping from input to label changes i.e., the same labeled corresponds to different data due to measurement differences.

⚙ Datasets and Types of Shifts



$$\mathcal{D}_S = \{(x_i^S, y_i^S)\}_{i=1}^{n_S}$$

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Label: spiral

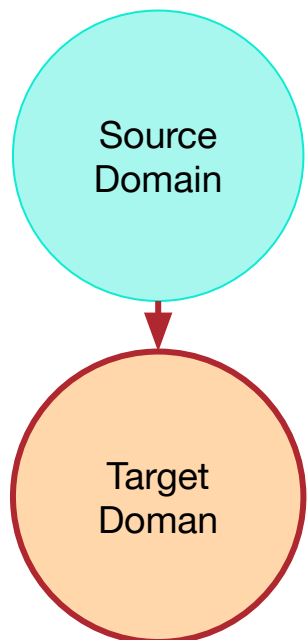
T: DESI
No Label

Covariate Shift: $P_s(X) \neq P_t(X)$ but $P_s(Y | X) = P_t(Y | X)$

Data distribution changes (often not just image properties but class distribution as well for example e.g., different sensors, sampling bias), but the conditional distribution of labels given the data remains the same.



Datasets and Types of Shifts



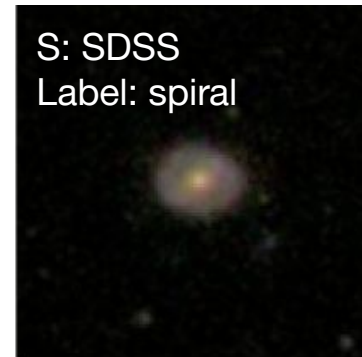
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with n_T unlabelled samples, from joint distribution $P_T(x)$

S: SDSS
Label: spiral



T: DESI
No Label



Compound Shift:

Any potential combination of two or more shift!



Dataset Shift Quiz !

Scenario 1:

You trained a model on a balanced dataset of merging and non-merging galaxies. At test time, your realistic catalog contains 90% of the non-merging galaxy images, and the remaining 10% are merging. The images for each galaxy still look the same.

Covariate

Concept / Conditional

Label

Answer:



Dataset Shift Quiz !

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Covariate

Concept / Conditional

Label

Answer:

Label shift — the label distribution $P(y)$ changed (mergers are rare), but image features ($P(x | y)$) stayed the same.



Dataset Shift Quiz !

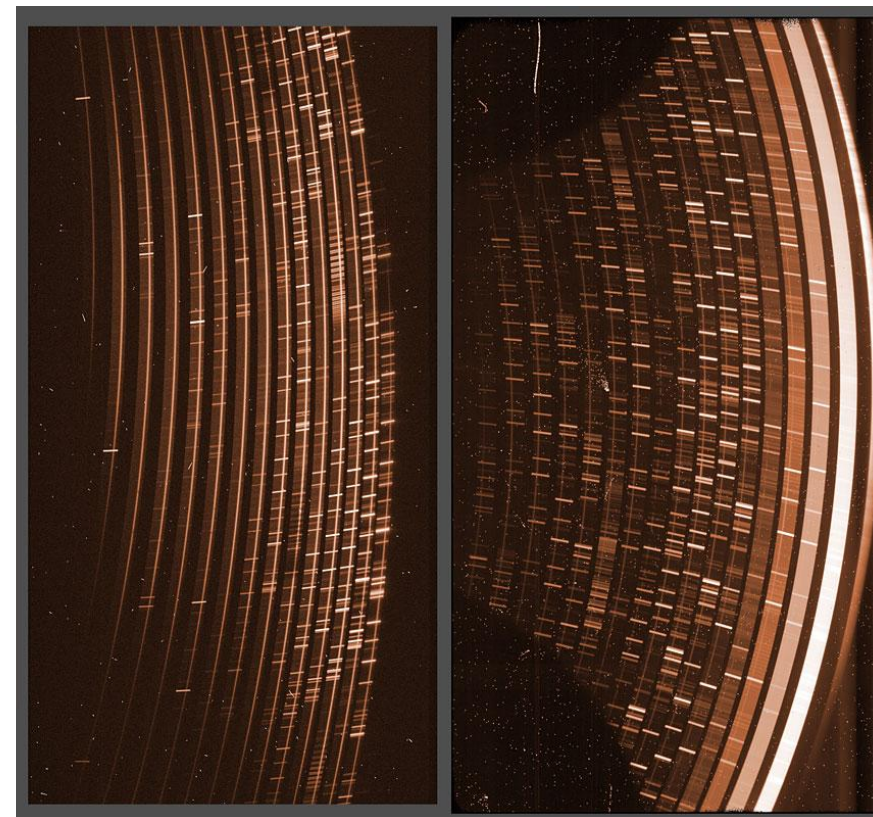
Scenario 2:

A spectral classification model is trained to identify star types using observations from the **VLT/X-Shooter** spectrograph in the Atacama Desert, Chile. Later, the model is applied to data from the **Keck/HIRES** spectrograph in Mauna Kea, Hawaii, where the star types and labels are the same, but atmospheric conditions, humidity levels, and telescope optics differ.

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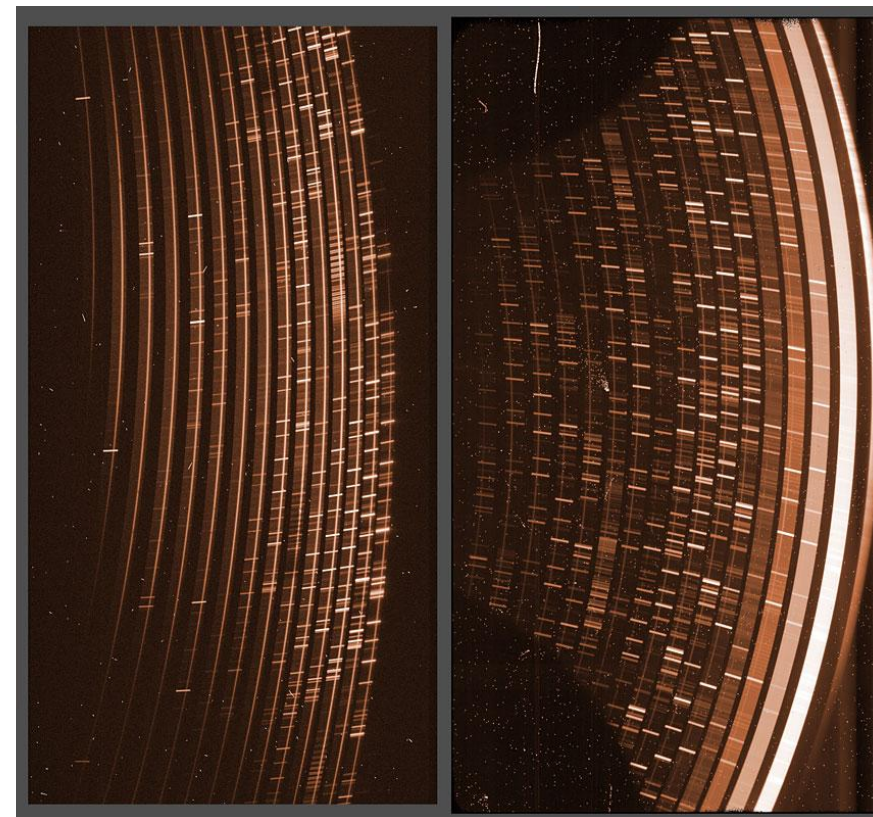
Covariate

Concept / Conditional

Label

Answer:

Covariate shift — the input distribution $\mathbf{P}(\mathbf{x})$ (stellar spectra features) has changed due to environmental factors, but $\mathbf{P}(\mathbf{y} | \mathbf{x})$ (how features map to star types) is assumed to remain the same.



Useful Resources:

- [“A Brief Review of Domain Adaptation” Farahini et al. 2020.](#)
- [“Deep Visual Domain Adaptation: A Survey” Wand & Deng 2018.](#)
- [“Domain Adaptation for Visual Applications: A Comprehensive Survey” Csurka 2017.](#)

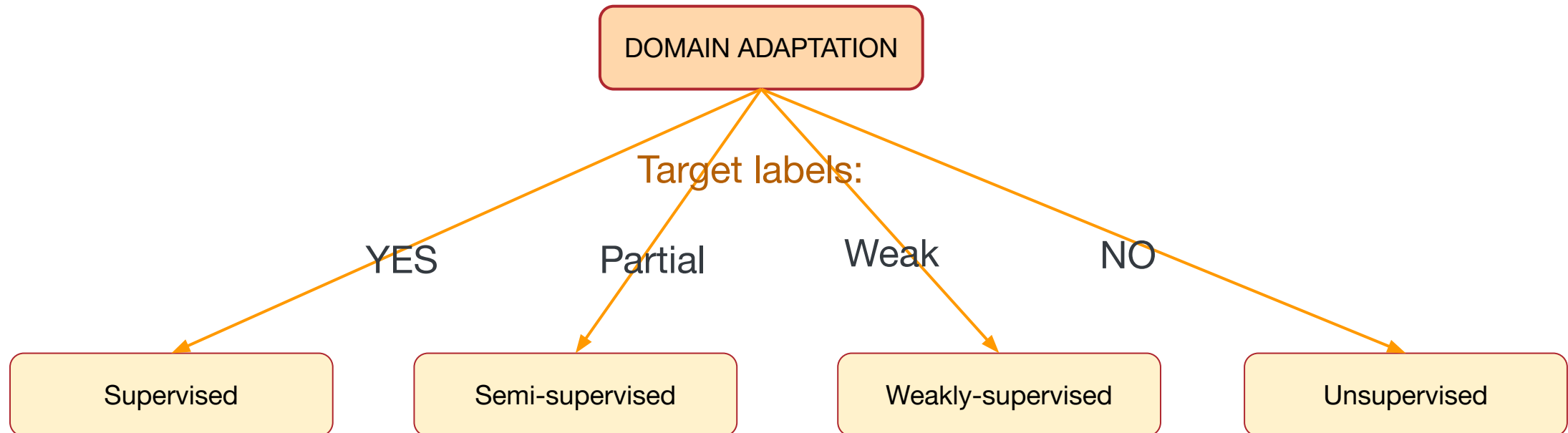
02

Domain Adaptation Techniques



Types of Domain Adaptation Approaches

There is a huge number of domain adaptation methods to suit different needs.

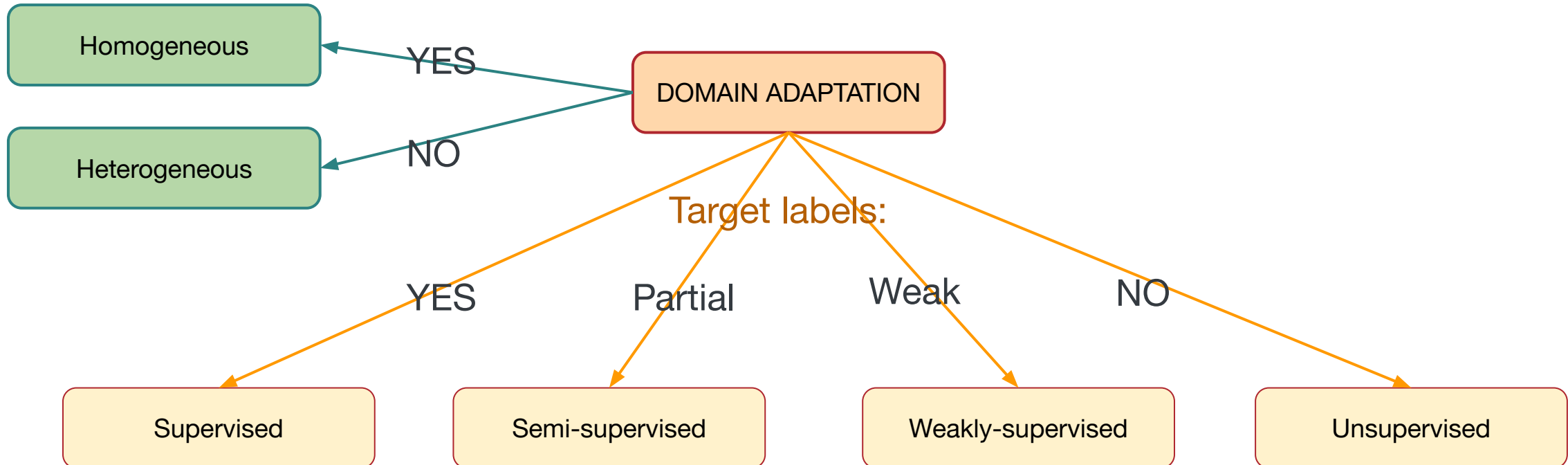




Types of Domain Adaptation Approaches

There is a huge number of domain adaptation methods to suit different needs.

Similarity of the
feature space:



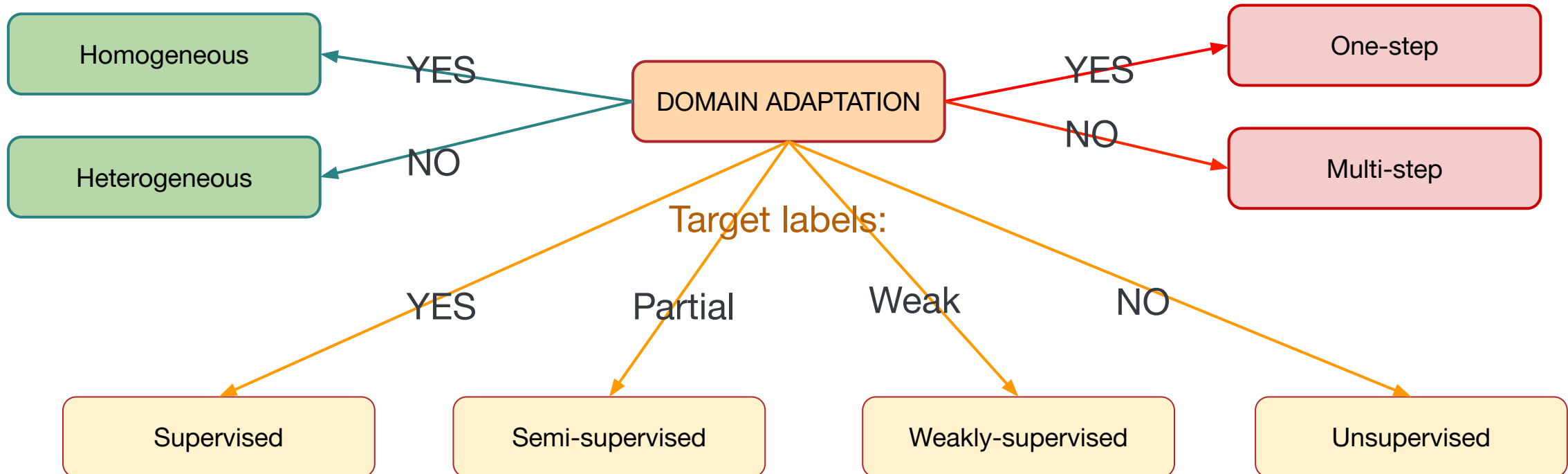


Types of Domain Adaptation Approaches

There is a huge number of domain adaptation methods to suit different needs.

Similarity of the
feature space:

Knowledge directly
transferred:





Types of Domain Adaptation Approaches

There is a huge number of domain adaptation methods to suit different needs.

Alignment approach:

Instance-based

Feature-based

reweight labeled data

find common features

Similarity of the
feature space:

Homogeneous

YES

Heterogeneous

NO

DOMAIN ADAPTATION

Knowledge directly
transferred:

YES

One-step

NO

Multi-step

Target labels:

YES

Supervised

Partial

Semi-supervised

Weak

Weakly-supervised

NO

Unsupervised

⚙️ Domain Adaptation Quiz !

Scenario 1:

You trained a cosmic ray classifier using labeled simulated data. You also have a small set of real cosmic ray events with noisy labels from an old detector. What is this setup best described as?

Supervised

Weakly supervised

Unsupervised

Answer:



⚙️ Domain Adaptation Quiz !

Scenario 1:

You trained a cosmic ray classifier using labeled simulated data. You also have a small set of real cosmic ray events with noisy labels from an old detector. What is this setup best described as?

Supervised

Weakly supervised

Unsupervised

Answer:

You have real data labels, but they are **noisy or imprecise** — so you need to use **weak supervision**. Supervised DA assumes accurate labels, while unsupervised DA assumes no labels. Weak supervision methods handle uncertainty, label correction, or probabilistic label modeling.



Domain Adaptation Quiz !

Scenario 2:

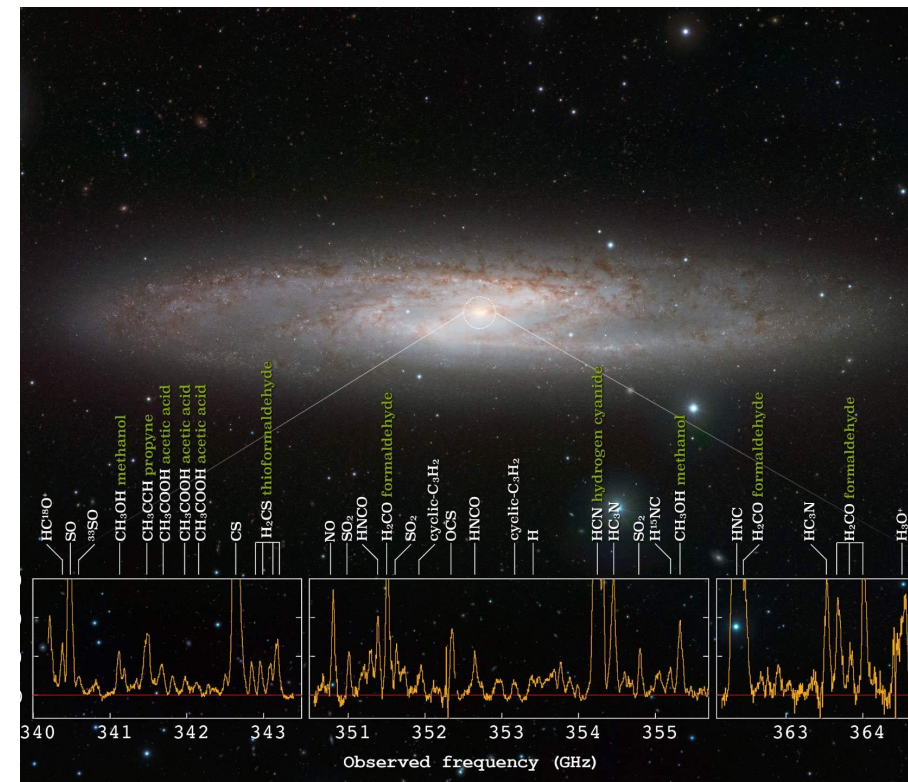
You trained a redshift estimation model using spectroscopic data (rich, high-dimensional spectra). You now want to apply it to photometric data (broadband magnitudes). What kind of domain adaptation challenge is this?

Unsupervised

Homogeneous

Heterogeneous

Answer:



Domain Adaptation Quiz !

Scenario 2:

You trained a redshift estimation model using spectroscopic data (rich, high-dimensional spectra). You now want to apply it to photometric data (broadband magnitudes). What kind of domain adaptation challenge is this?

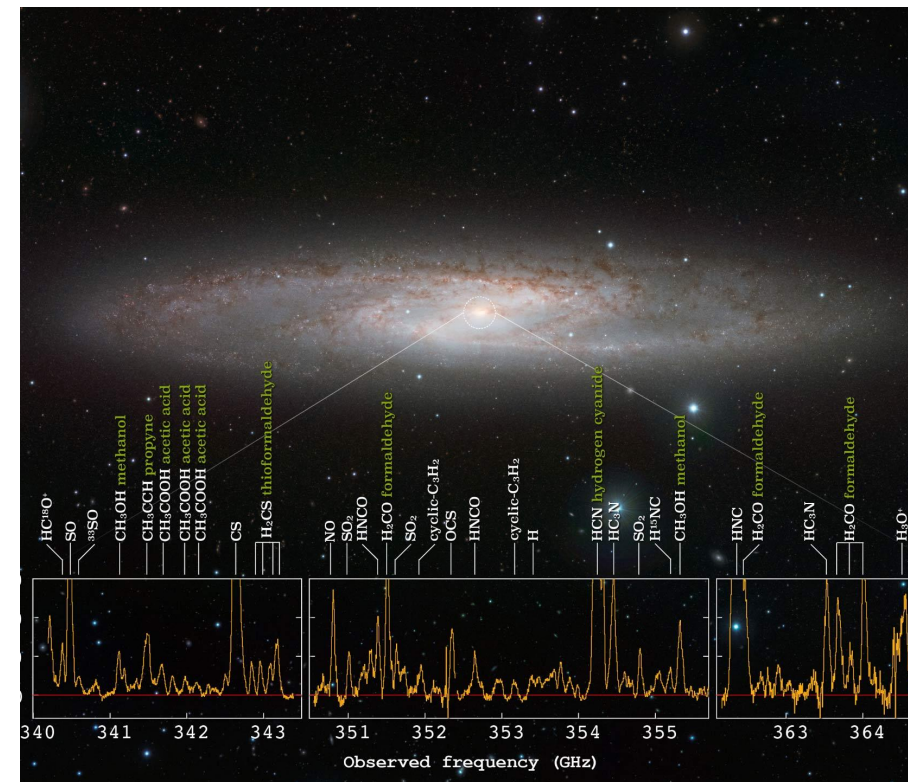
Unsupervised

Homogeneous

Heterogeneous

Answer:

The **input features differ in type and dimension** — spectra vs. photometry — so this is a **heterogeneous DA** problem. Homogeneous DA assumes the same input space. The key challenge here lies in mapping between incompatible feature spaces (e.g., using shared latent representations or cross-modal alignment).





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Feature-based

reweight labeled data

find common features

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DOMAIN ADAPTATION

Knowledge directly
transferred:

One-step

YES

NO

Multi-step

Target labels:

YES

Partial

Weak

NO

Supervised

Semi-supervised

Weakly-supervised

Unsupervised



Combining Datasets

DOMAIN ADAPTATION

Align data distributions in the latent space of the network by forcing the network to **find more robust domain-invariant features**.

Combining Datasets

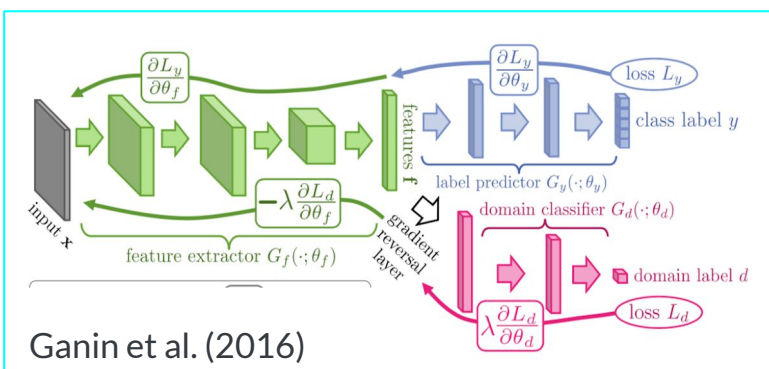
DOMAIN ADAPTATION

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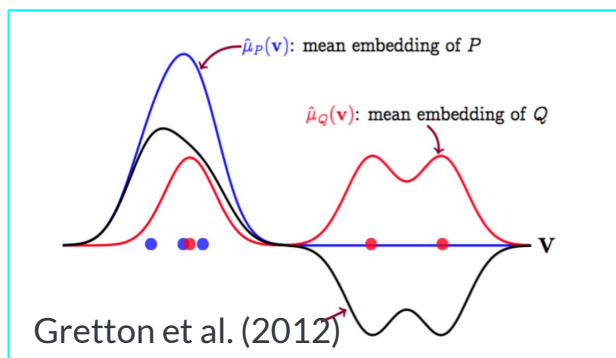
Adversarial

Distance-based
methods

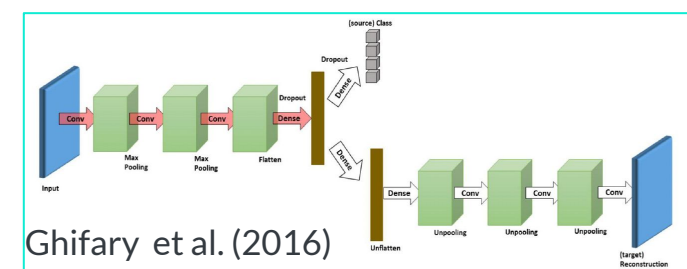
Reconstruction-based



Using domain discriminators to encourage domain confusion through an adversarial objective.



Minimize a distance metric between source and target data.



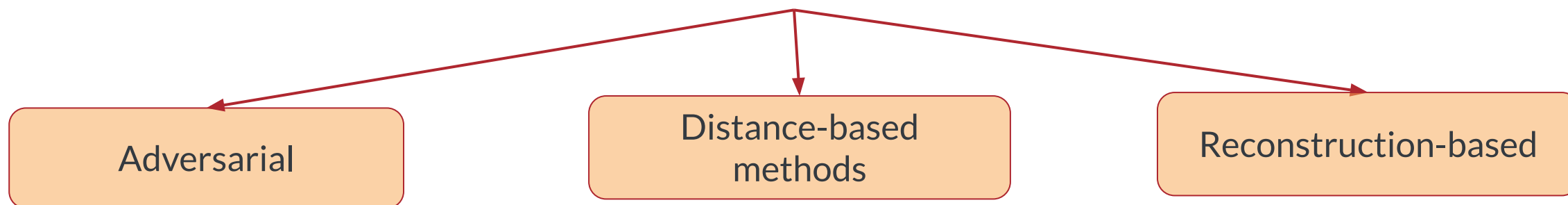
Data reconstruction as an auxiliary task to ensure feature invariance.



Combining Datasets

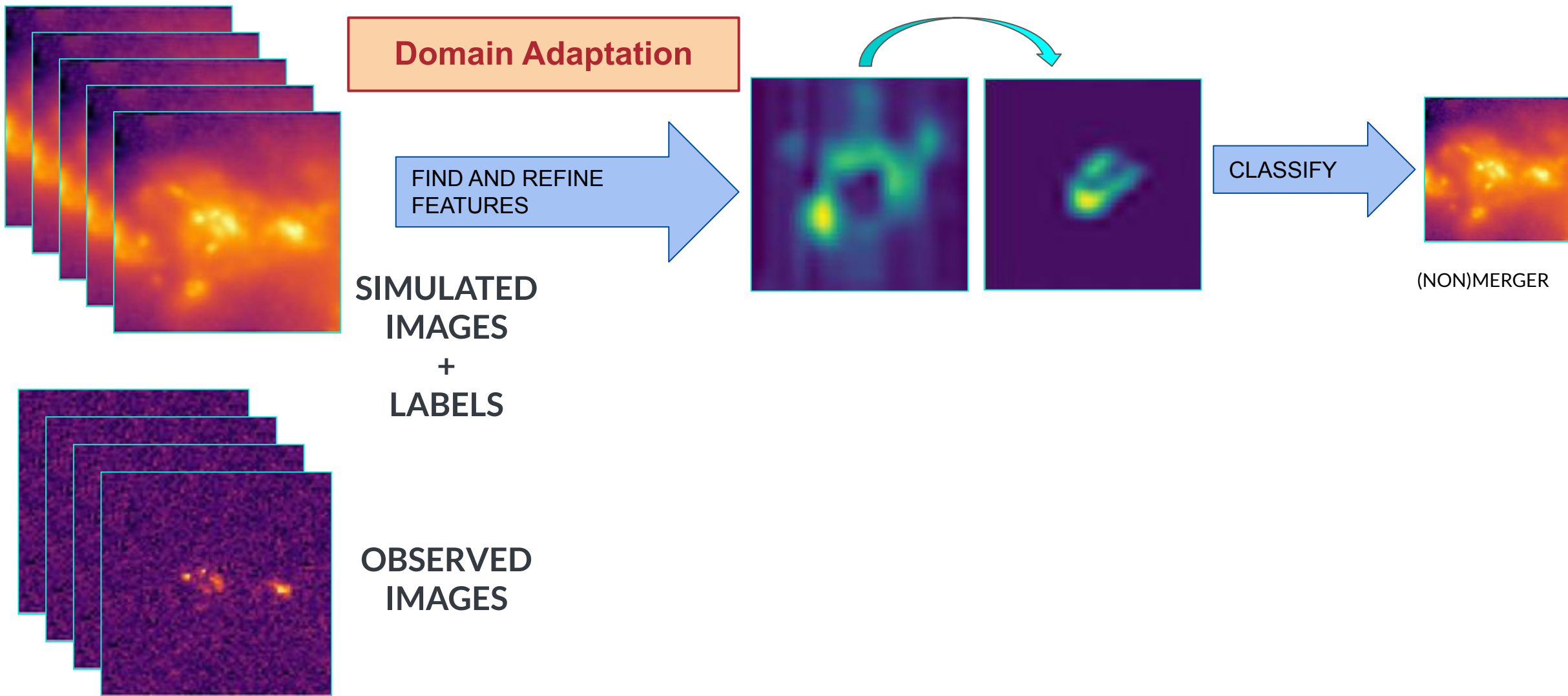
DOMAIN ADAPTATION

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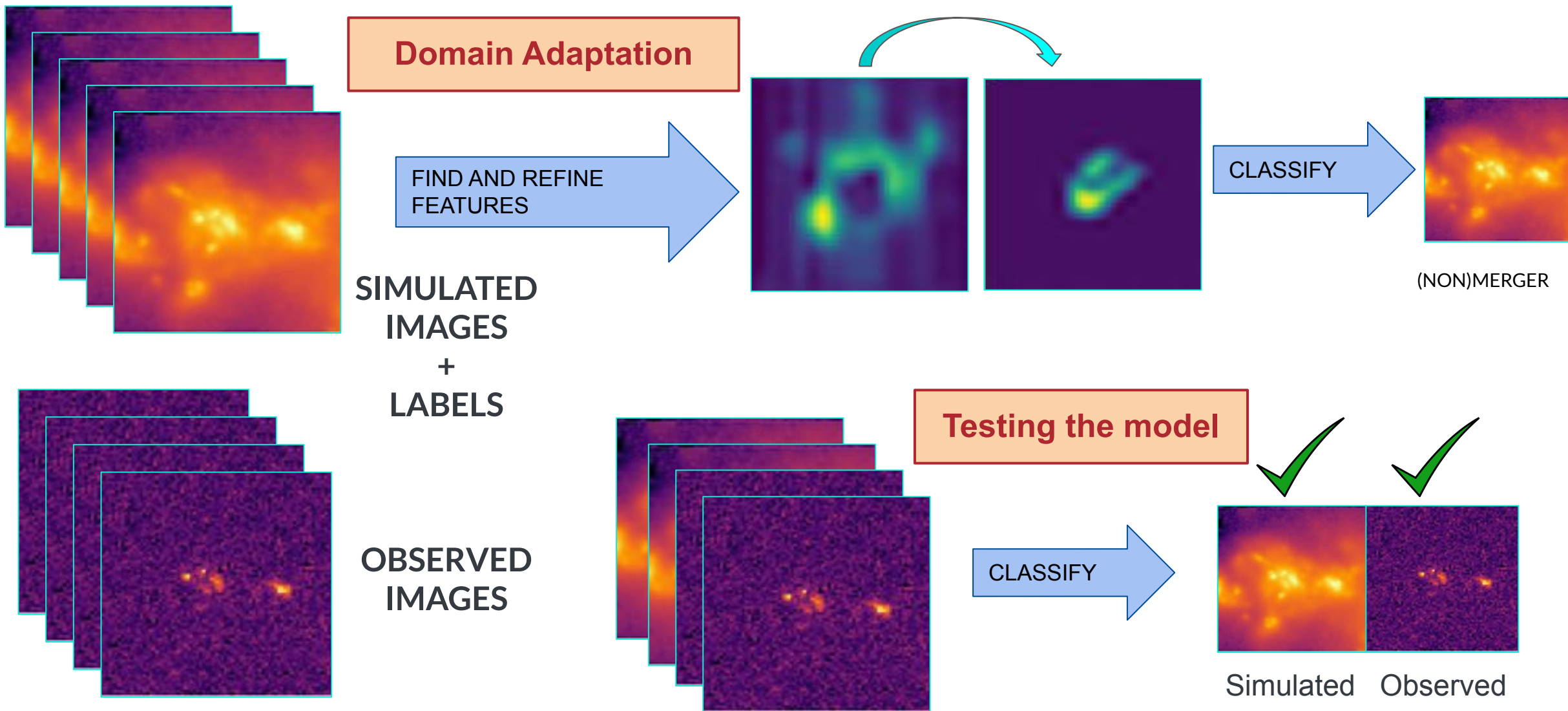


Works on **unlabeled target domain!**
Can be applied to **new data**, no need for scientists to label anything.

Training with Domain Adaptation



Training with Domain Adaptation



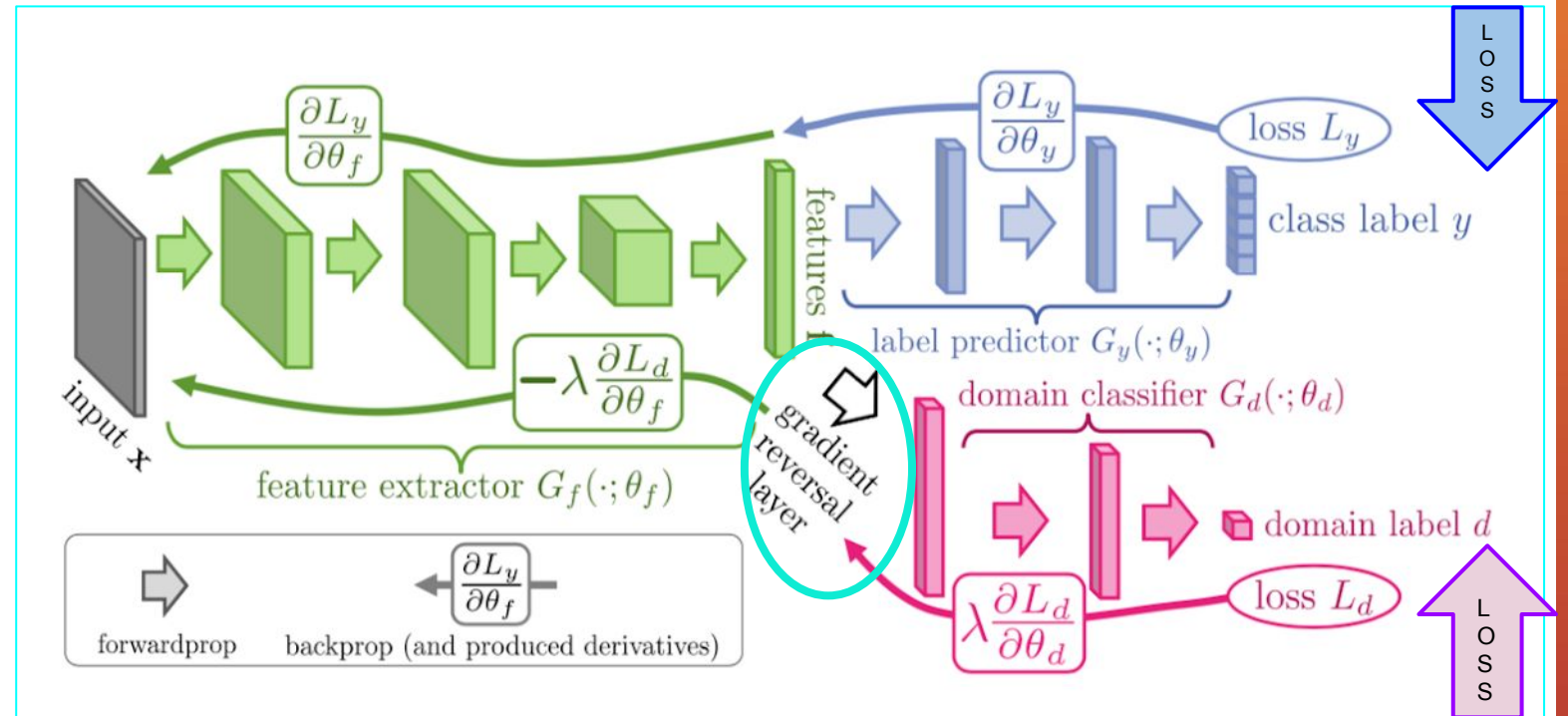


Adversarial Training

Domain Adversarial Neural Networks - DANNs

DANN - feature extractor + label predictor + domain classifier

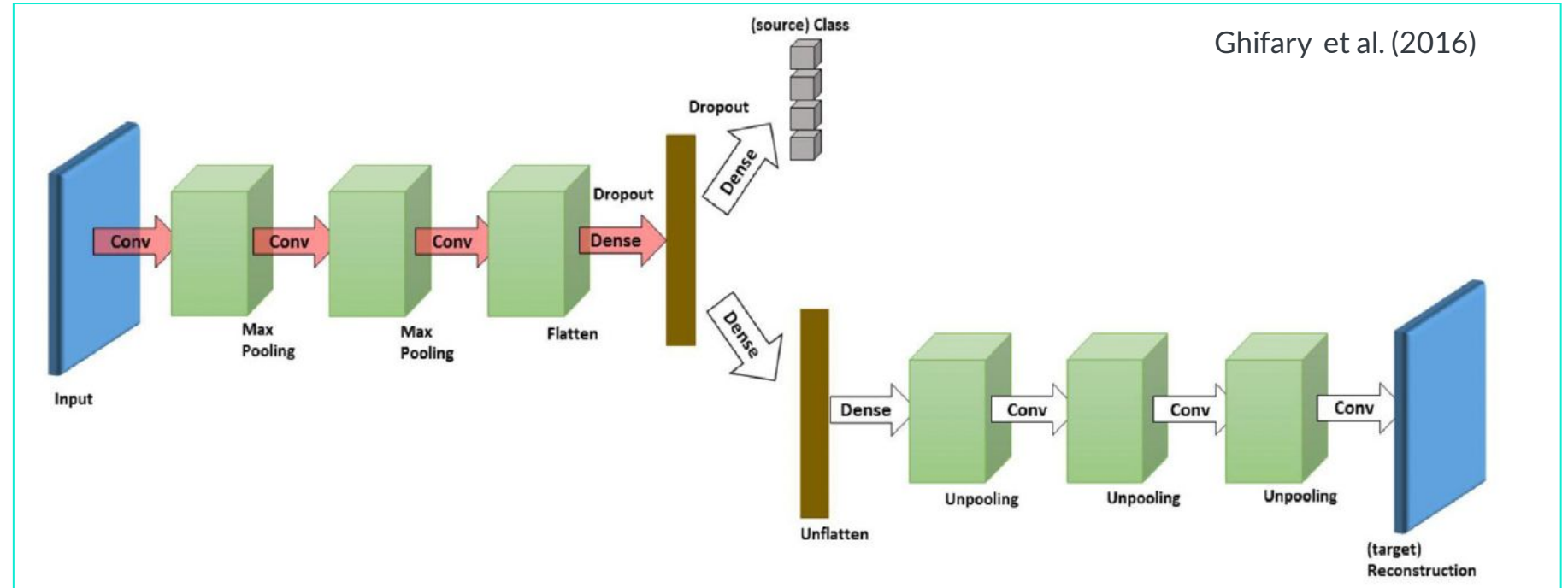
- **Gradient reversal layer** - multiplies the gradient by a negative constant during the backpropagation.
- Results in the extraction of **domain-invariant features**.
- Only source domain images are labeled during training.



Ganin et al. (2016)

Reconstruction-Based Training

Deep Reconstruction-Classification Networks - DRCN



Model jointly learns a shared encoding representation for two tasks:

- **supervised classification** of labeled source data
- **unsupervised reconstruction** of unlabeled target data.

The learnt representation not only preserves discriminability, but also encodes useful information from the target domain.

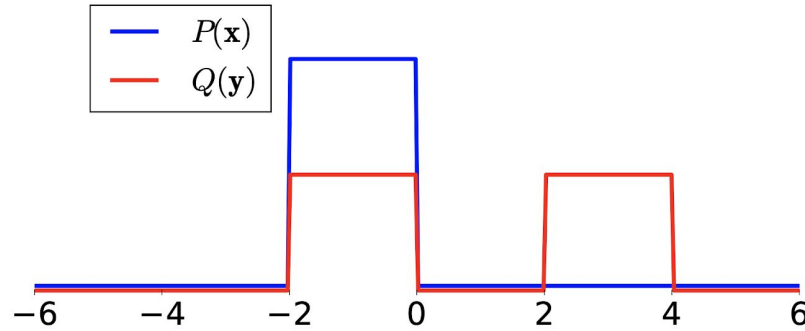


Distance-Based Training

Maximum Mean Discrepancy - MMD

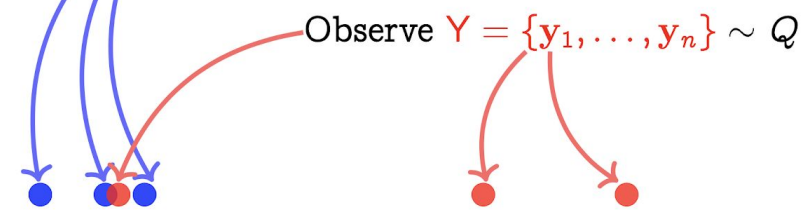
Smola et al. (2007)
Gretton et al. (2012)

Are P and Q different?



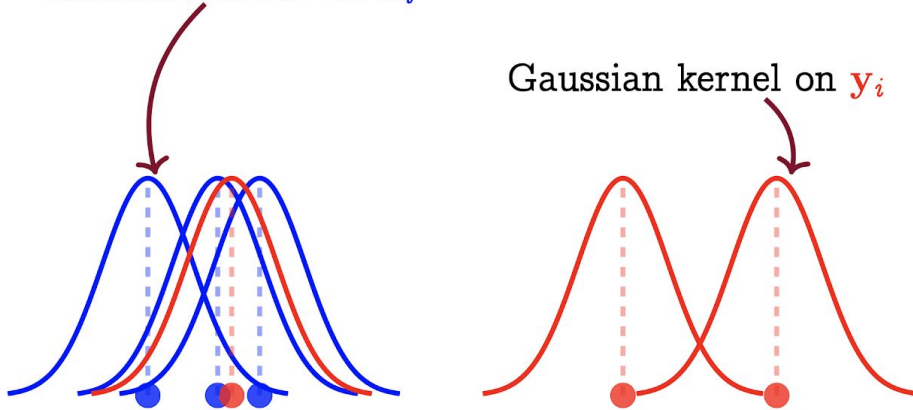
Observe $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \sim P$

Observe $Y = \{\mathbf{y}_1, \dots, \mathbf{y}_n\} \sim Q$



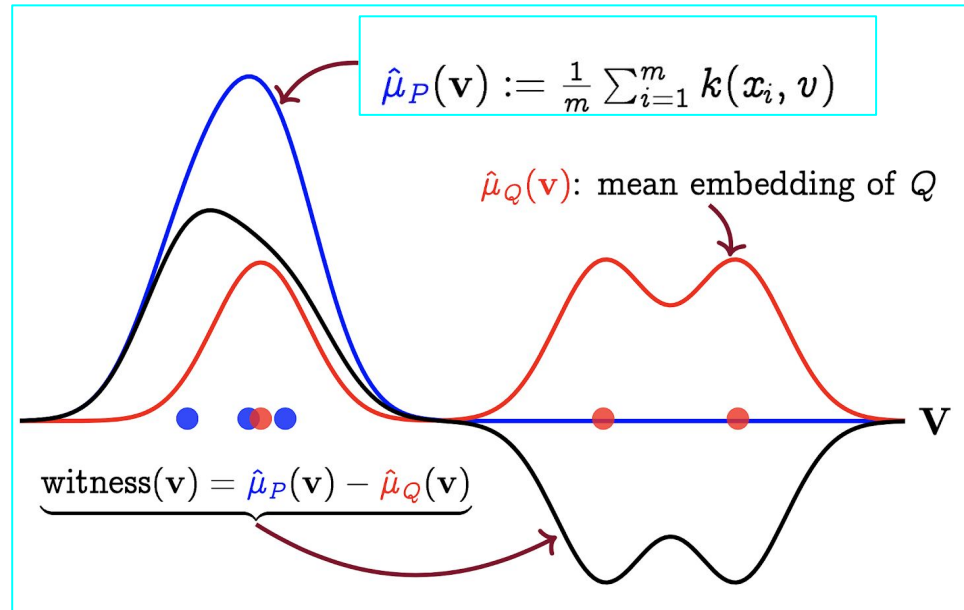
Gaussian kernel on $\mathbf{x}_i$

Gaussian kernel on $\mathbf{y}_i$



$$\hat{\mu}_P(\mathbf{v}) := \frac{1}{m} \sum_{i=1}^m k(\mathbf{x}_i, \mathbf{v})$$

$\hat{\mu}_Q(\mathbf{v})$: mean embedding of Q

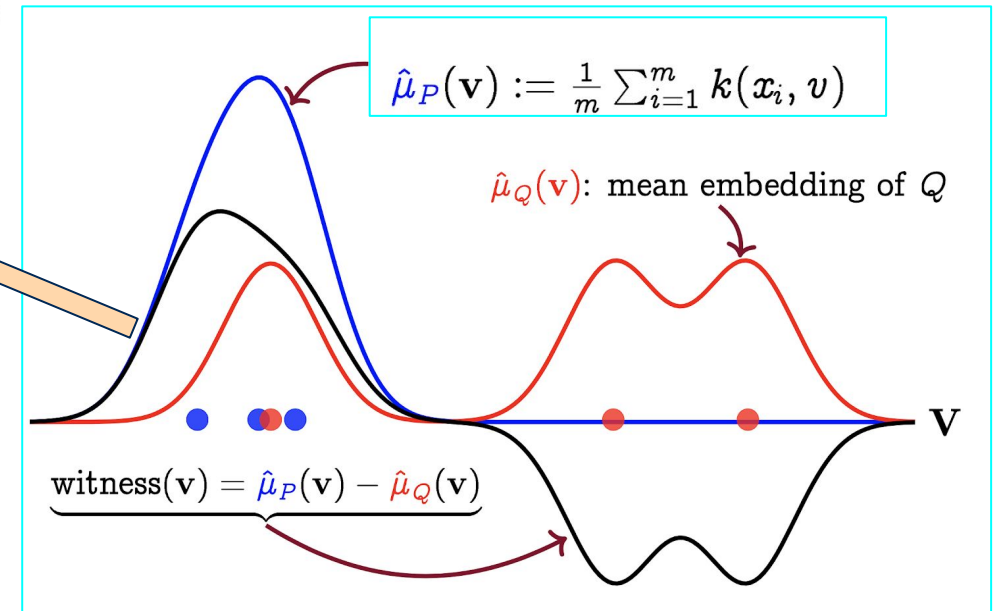
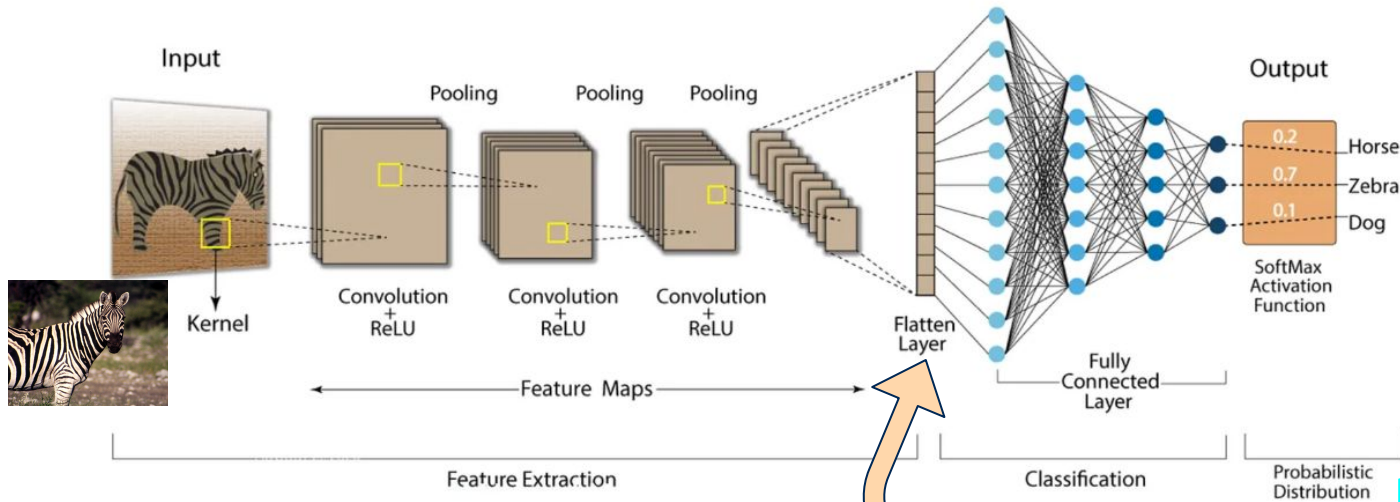




Distance-Based Training

Maximum Mean Discrepancy - MMD

Smola et al. (2007)
Gretton et al. (2012)

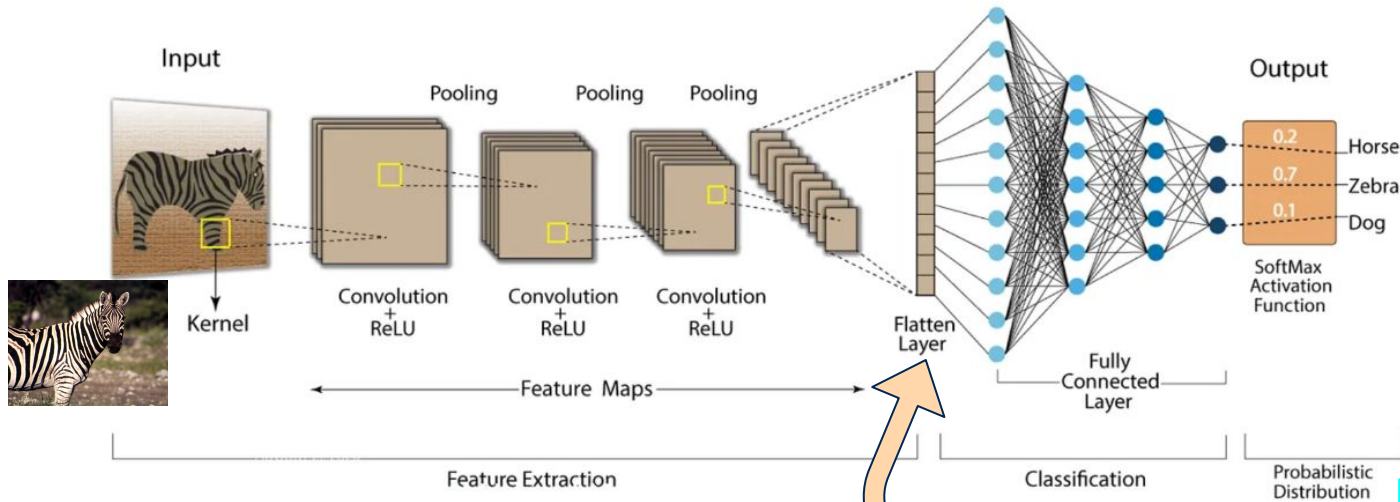




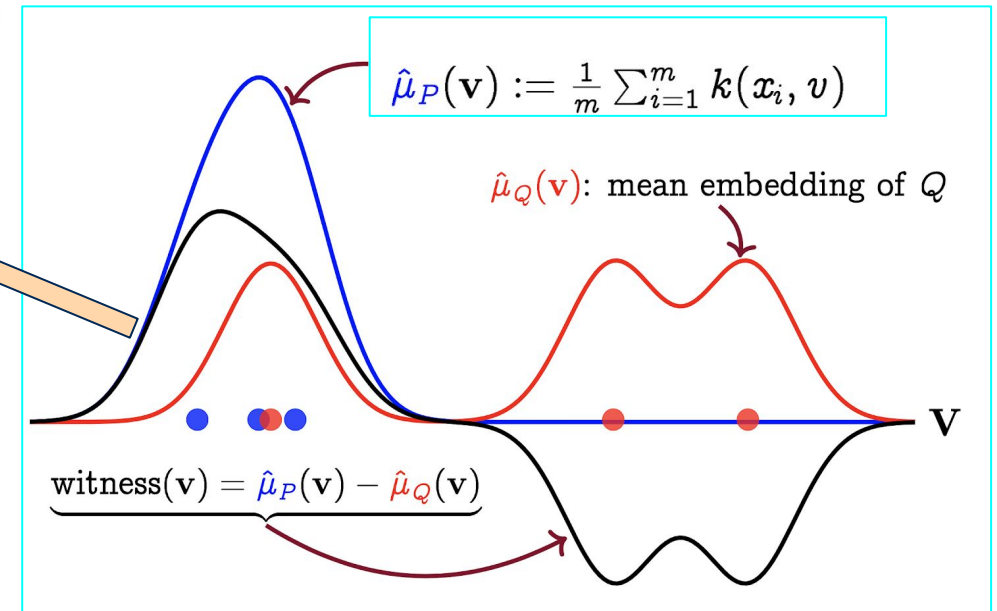
Distance-Based Training

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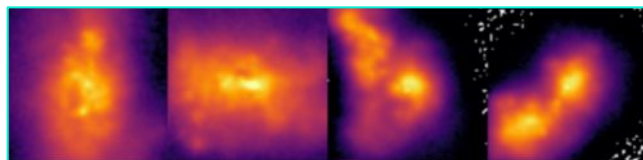


Domain adaptation method of your choice is applied to the latent embedding.
What happens then?...Stay tuned!

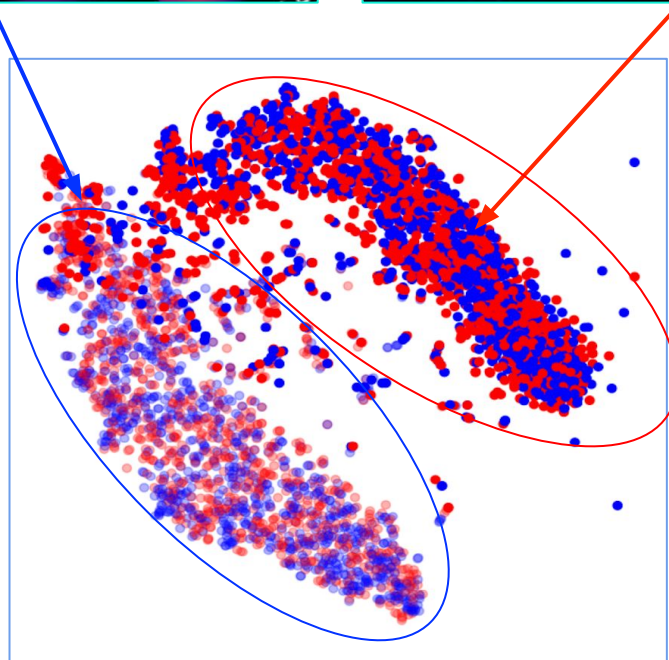
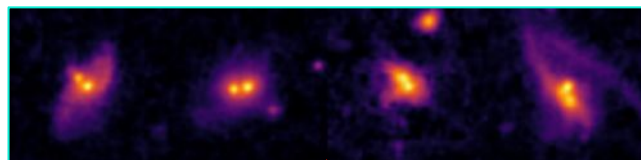


Latent Space Alignment

Source - Illustris



Target - SDSS observations

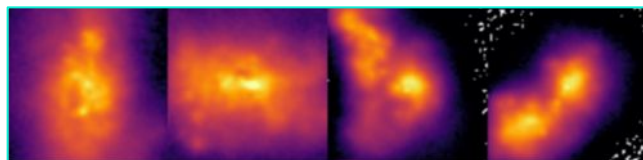


← This is how the network sees the data.
2D representation of network's latent space.

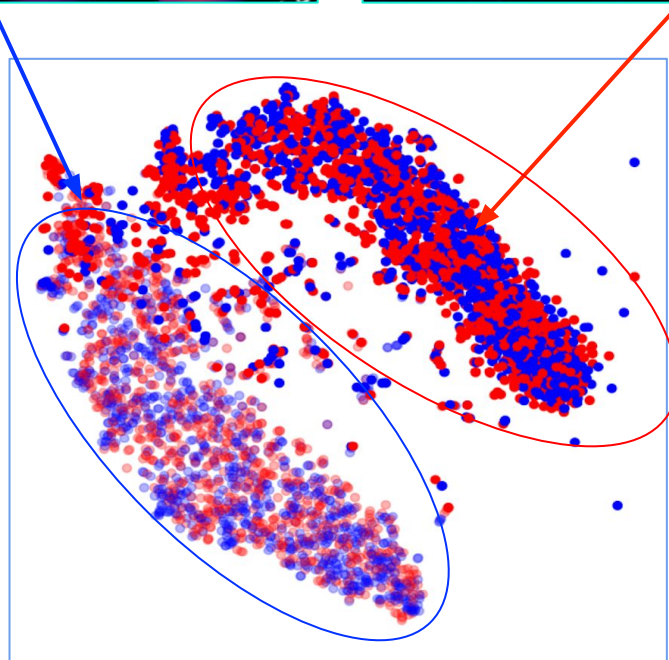
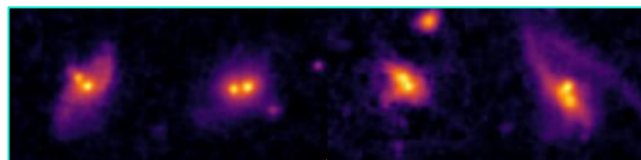
Ćiprijanović et al. 2020.
Ćiprijanović et al. 2021.

⚙️ Latent Space Alignment

Source - Illustris



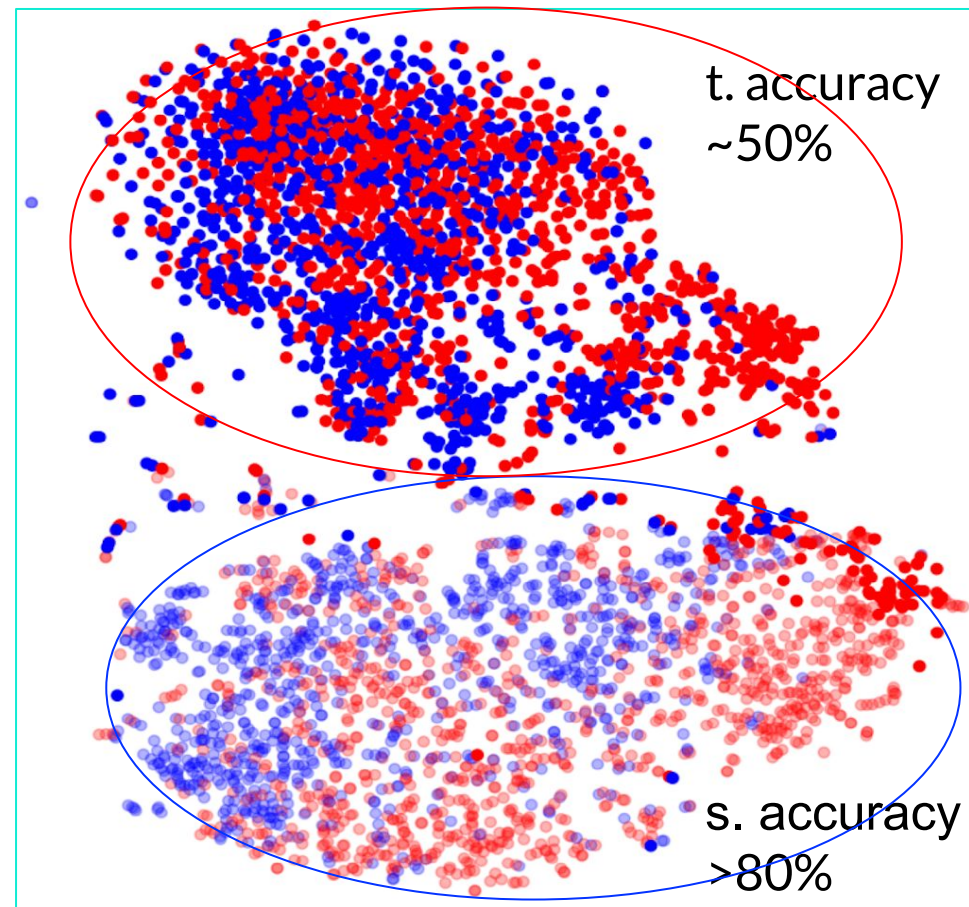
Target - SDSS observations



Ćiprijanović et al. 2020.
Ćiprijanović et al. 2021.



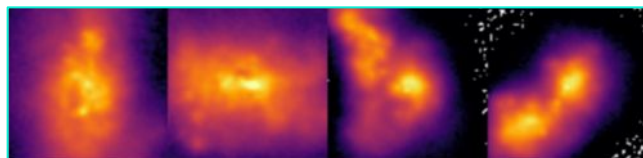
Regular Training



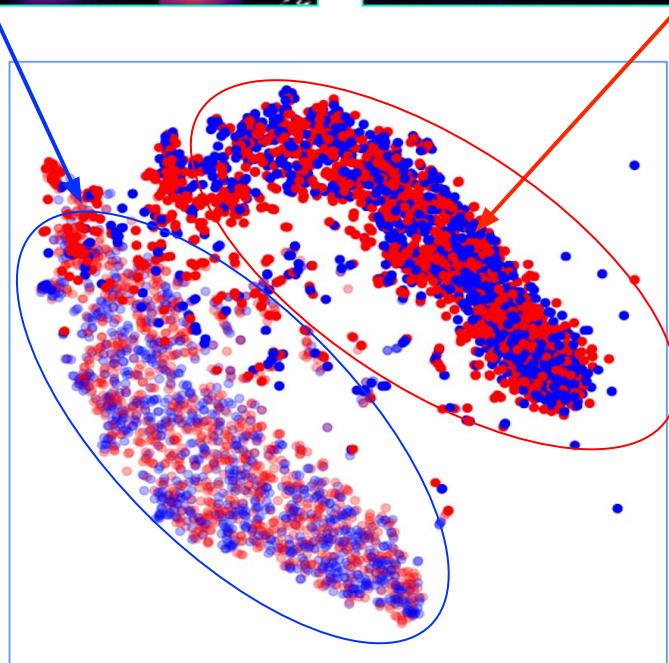
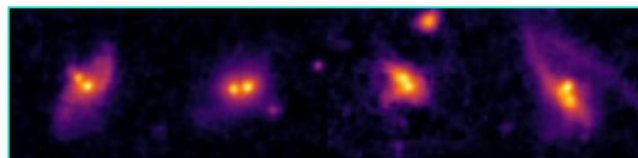


Latent Space Alignment

Source - Illustris



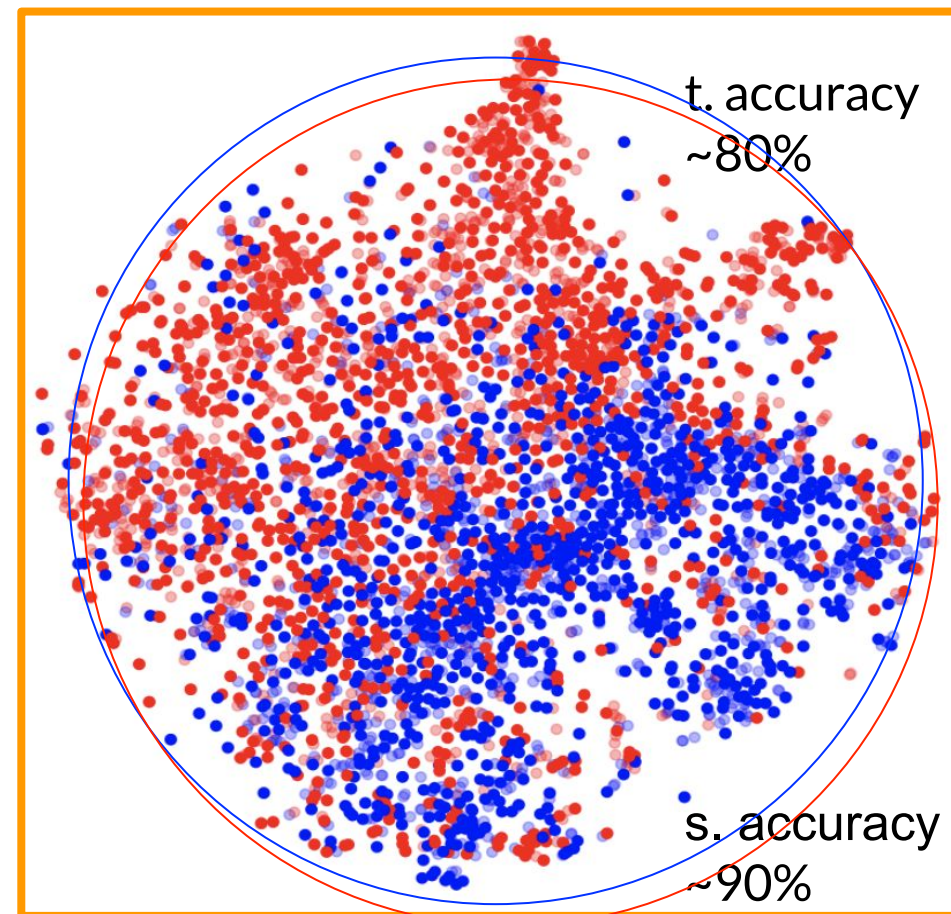
Target - SDSS observations



Ćiprijanović et al. 2020.
Ćiprijanović et al. 2021.

Domain Adaptation

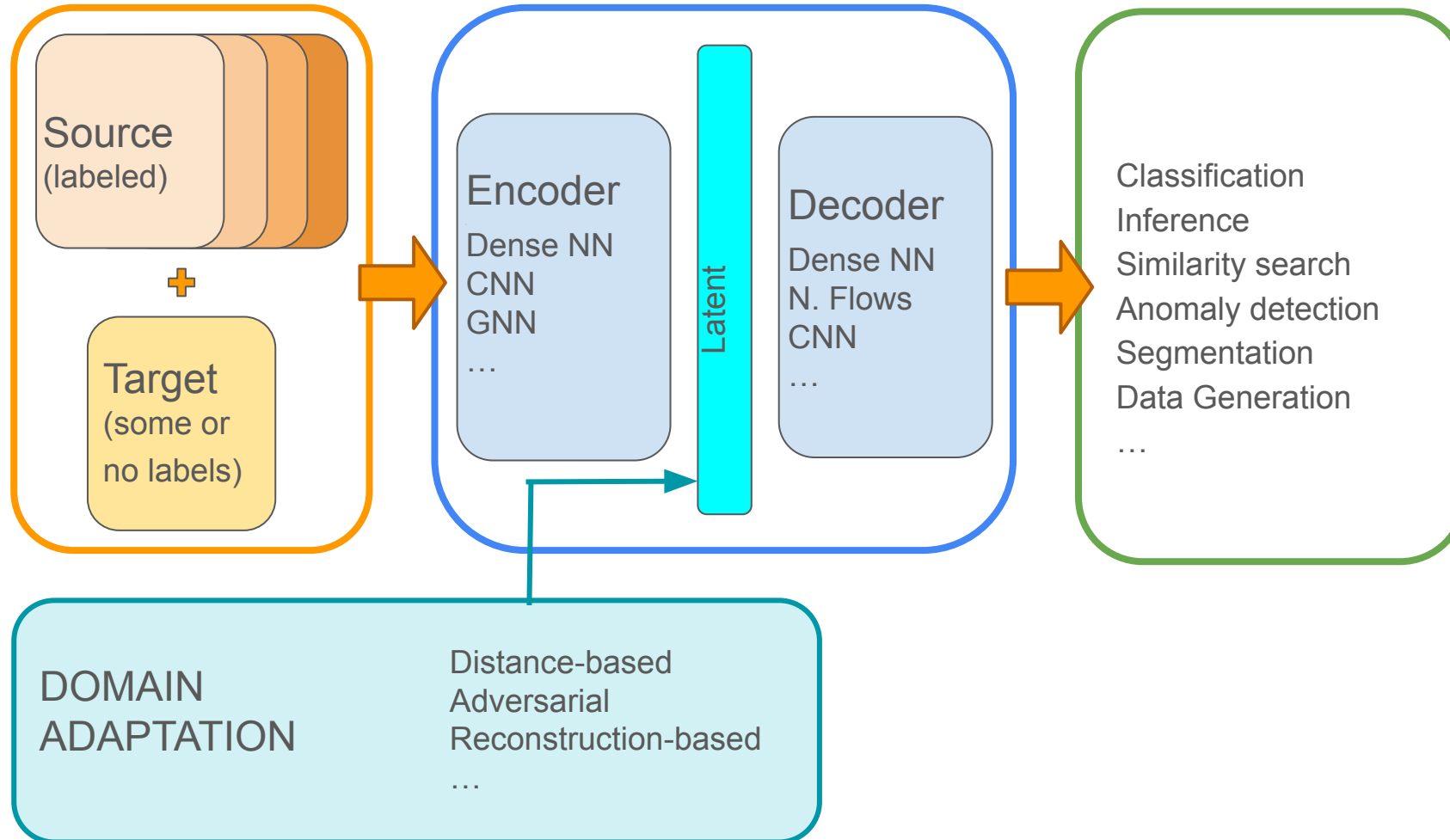
Up to 30% increase!





Where can we use it?

Any data, any task, any problem (within reason!)



03

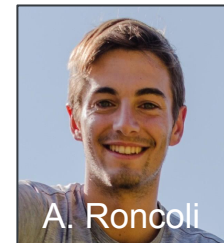
Examples

Useful Resources:

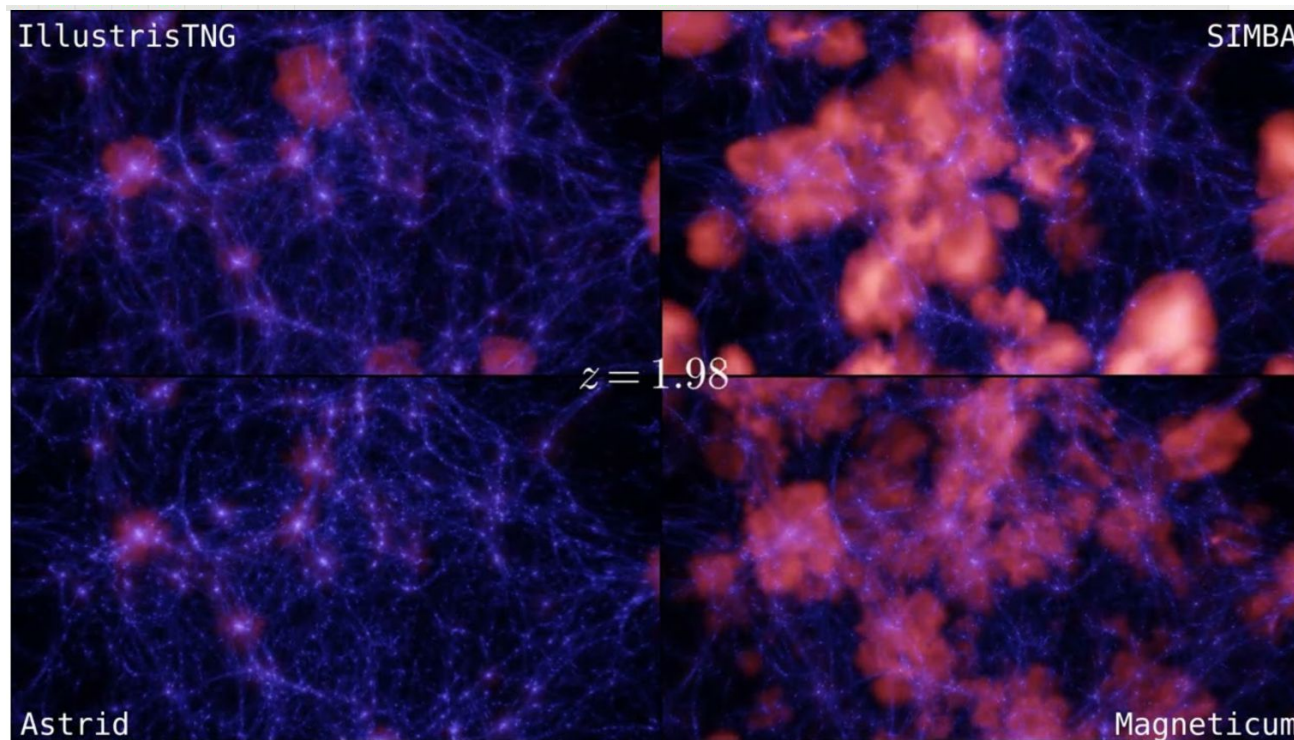
- [“SIDDA: Sinkhorn Dynamic Domain Adaptation for Image Classification with Equivariant Neural Networks” Pandya et al. 2025.](#)
- [“Domain-Adaptive Neural Posterior Estimation for Strong Gravitational Lens Analysis” Swierc et al. 2024.](#)
- [“Domain Adaptive Graph Neural Networks for Constraining Cosmological Parameters Across Multiple Data Sets” Roncoli et al. 2023.](#)

Cosmology with Graphs

NeurIPS 2023.
Roncoli et al. 2023.



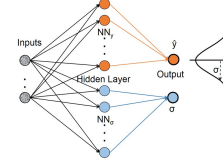
Can we correctly predict cosmology across different cosmological simulations?





Cosmology with Graphs

MVE: Infer matter density cosmological parameter

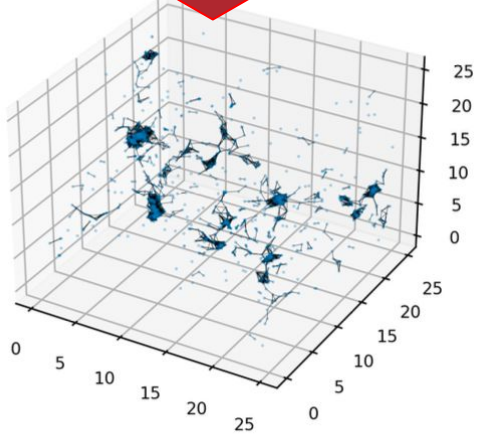
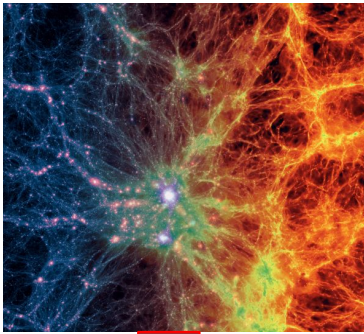


NeurIPS 2023.
Roncoli et al. 2023.



A. Roncoli

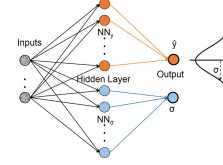
Graph Neural Networks:
ideal for sparse
galaxy catalogs!





Cosmology with Graphs

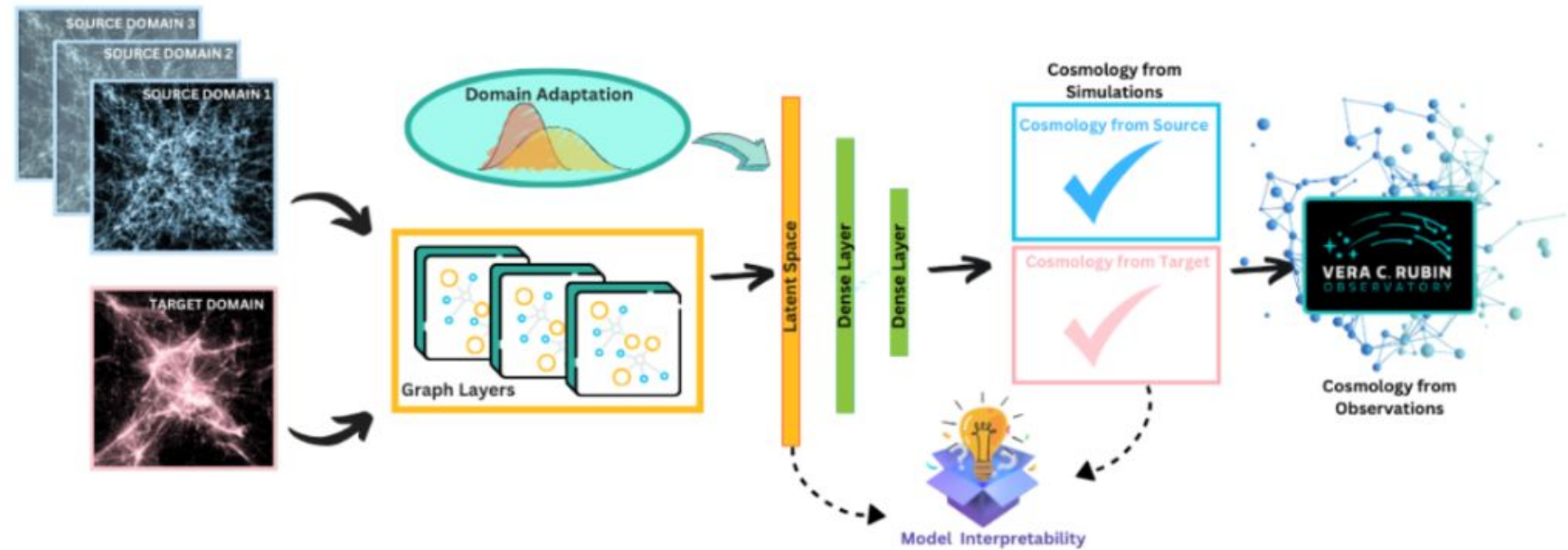
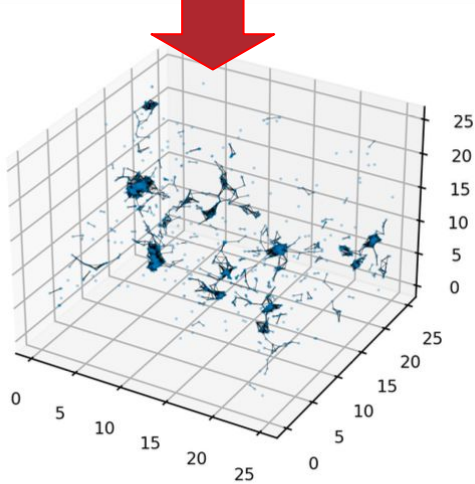
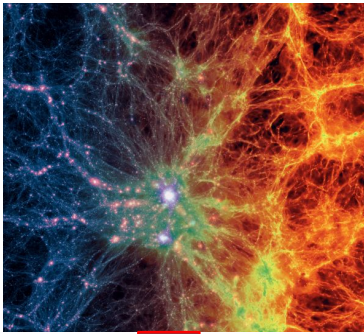
MVE: Infer matter density cosmological parameter



NeurIPS 2023.
Roncoli et al. 2023.

A. Roncoli

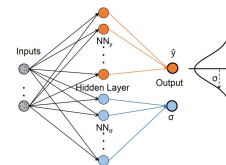
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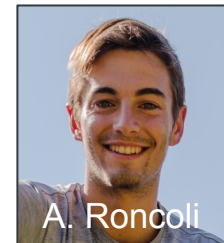


Cosmology with Graphs

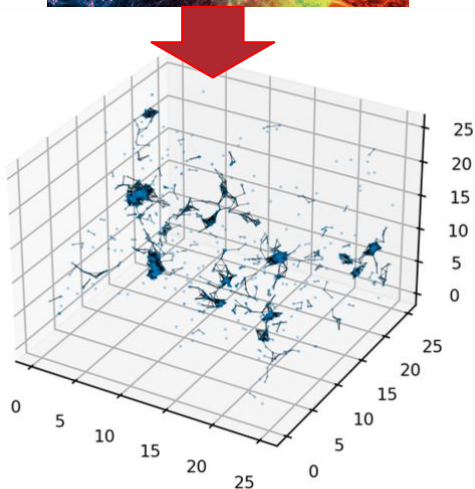
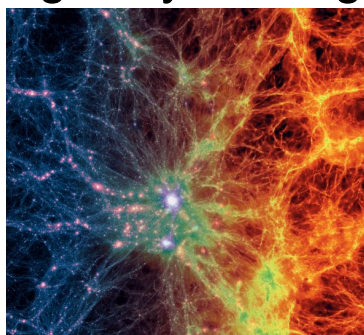
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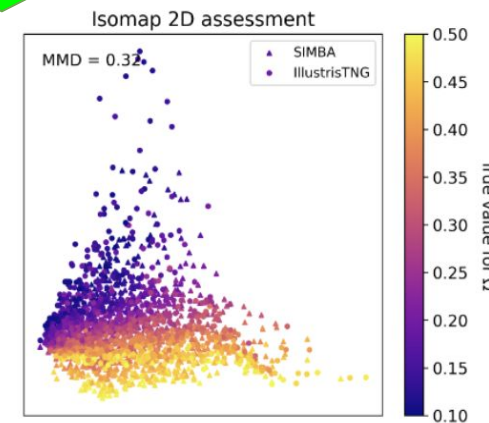
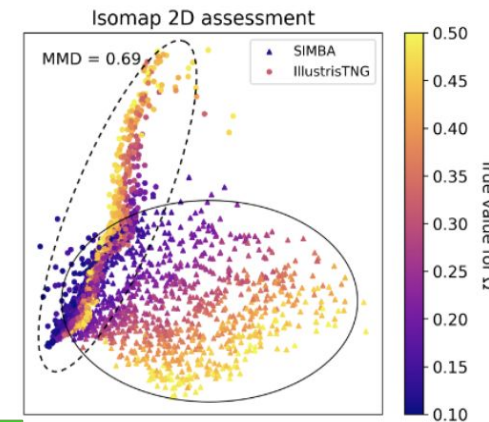
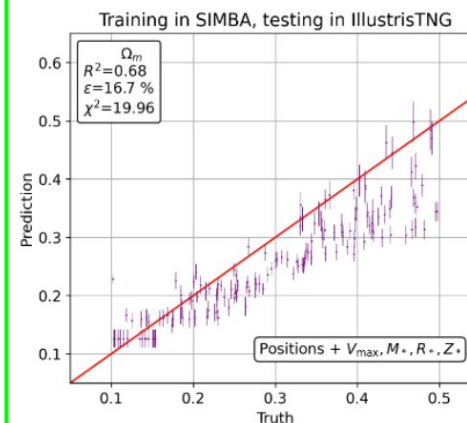
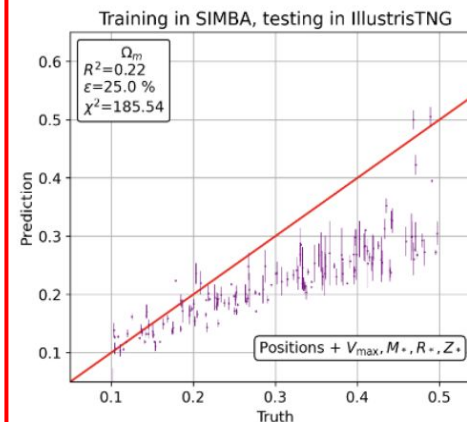
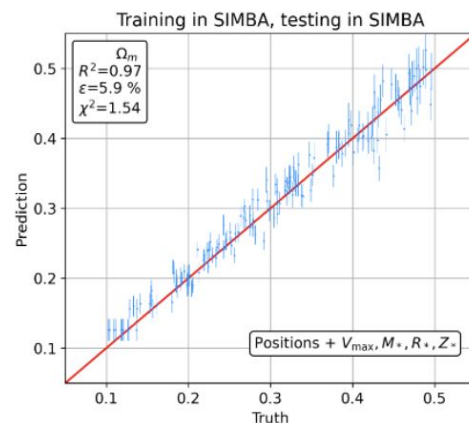
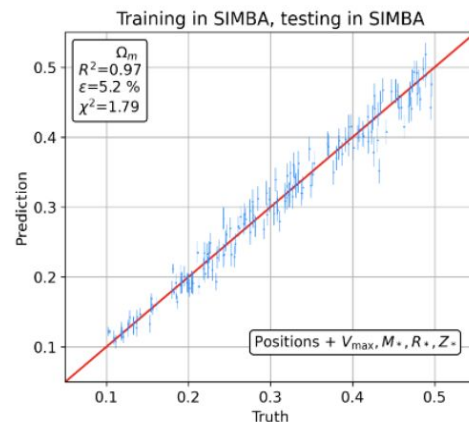
NeurIPS 2023.
Roncoli et al. 2023.



Graph Neural Networks:
ideal for sparse
galaxy catalogs!



z=0 1000 simulations each



**28%
better
relative
error**

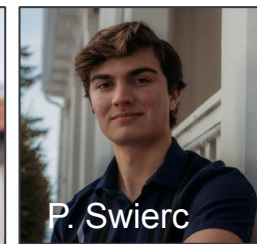
**Order
of mag.
better
 χ^2**



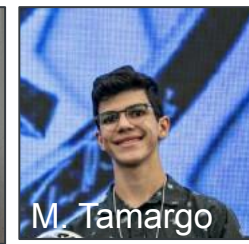
Strong gravitational lensing



S. Agarwal

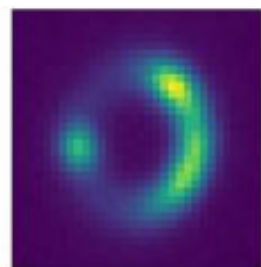


P. Swierc

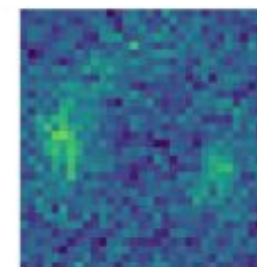
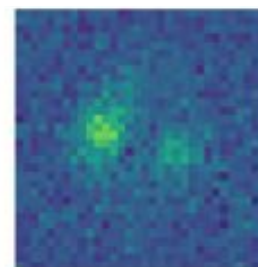
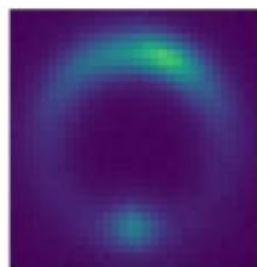


M. Tamargo

Can we infer strong lensing parameters (Einstein radius, ellipticity, position/offset) robustly in both simulated and real data?



Source - Clean

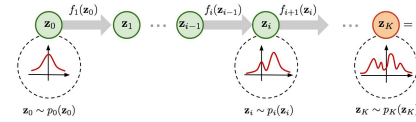


Target - DES



Strong gravitational lensing

SBI - infer Einstein radius, ellipticity, position/offset



S. Agarwal



P. Swierc

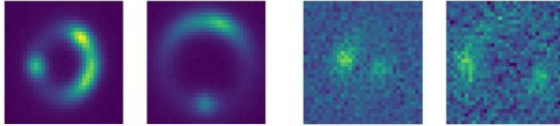


M. Tamargo

(a)

Source

Target



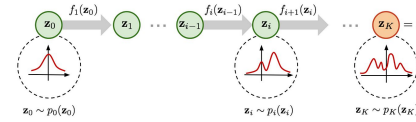
- a. 100k images (1-filter) with and without noise
- b. **Latent distributions are correctly aligned.**
- c. Differences between true values and predicted posteriors are small. **Accuracy improves up to 2 orders of magnitude.**
- d. Posterior are well calibrated and no longer overconfident.

Swierc et al. 2024.



Strong gravitational lensing

SBI - infer Einstein radius, ellipticity, position/offset



S. Agarwal

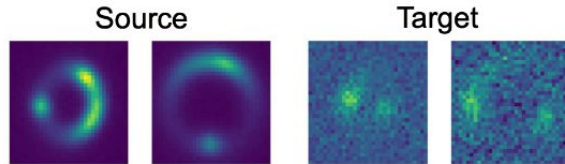


P. Swierc

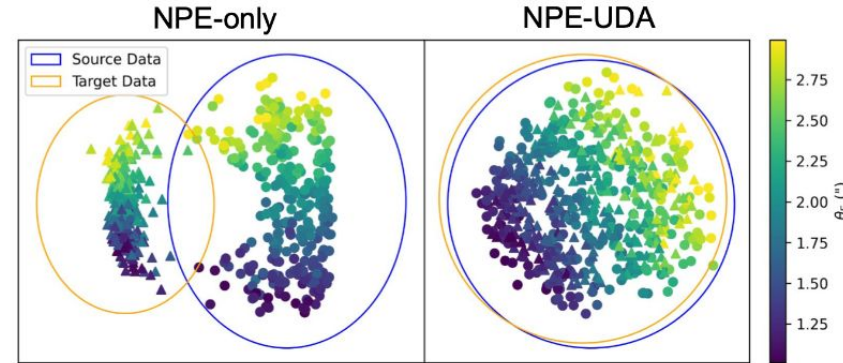


M. Tamargo

(a)



(b)



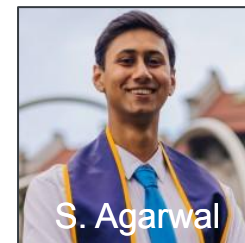
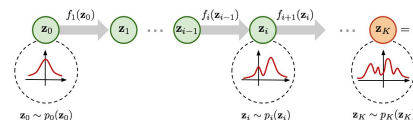
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Swierc et al. 2024.

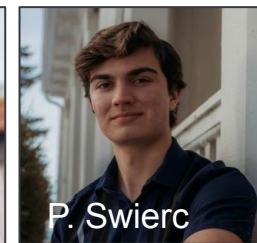


Strong gravitational lensing

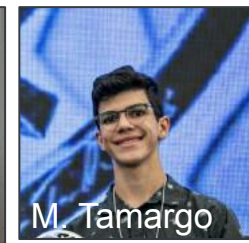
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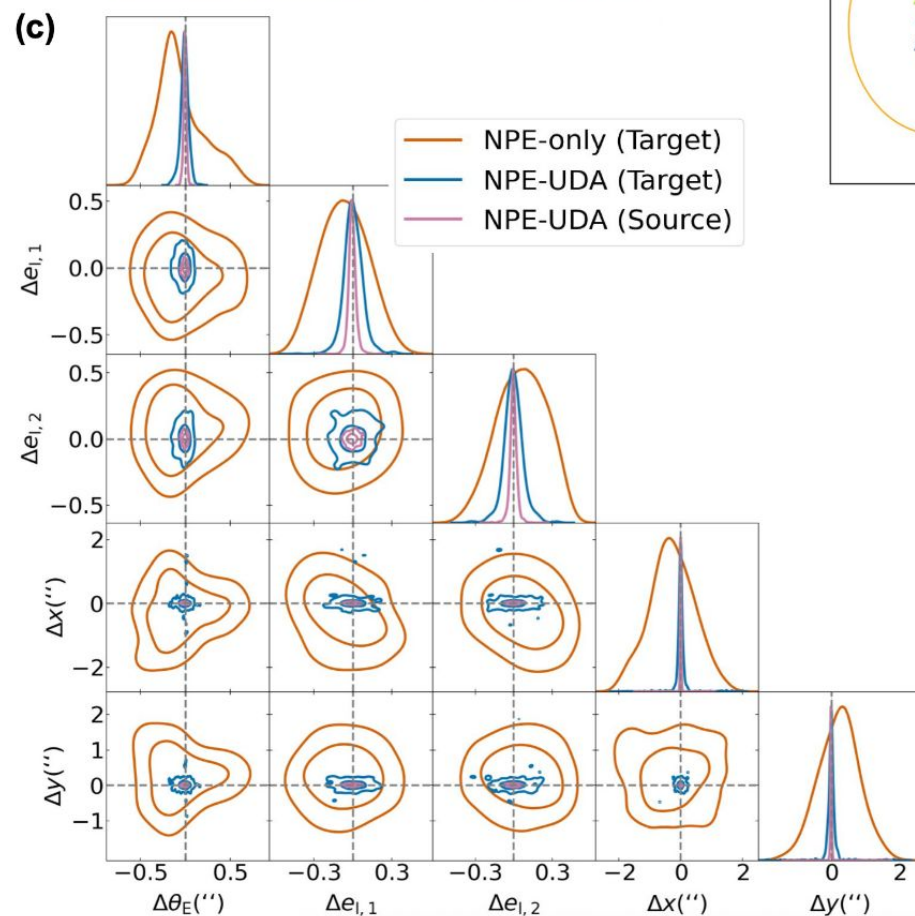
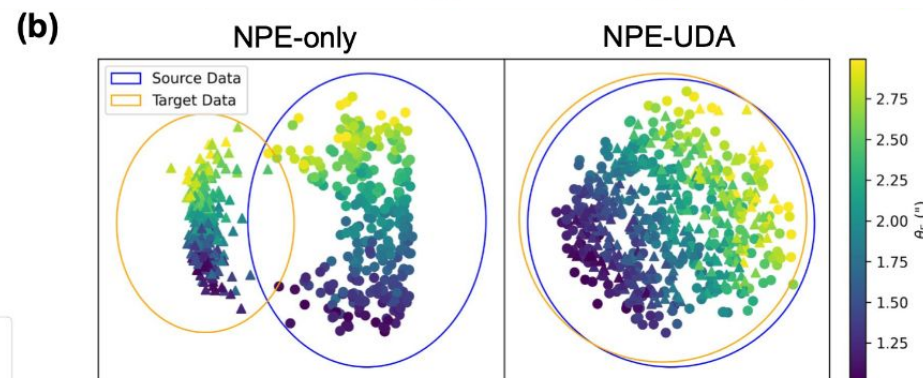
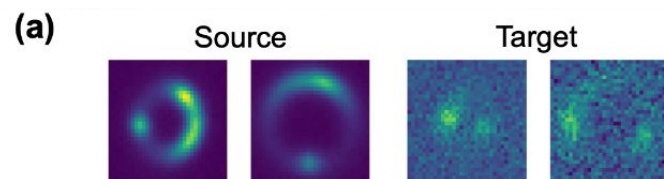
S. Agarwal



P. Swierc



M. Tamargo



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Swierc et al. 2024.



04

Open Questions & Recent Trends

Trending Topics



1. Unsupervised and Few-shot DA

- Moving beyond labeled source-heavy settings.
- Combining **few-shot learning** and **DA** to adapt with just a few target labels.

2. Test-time Adaptation (TTA) & Source-free Domain Adaptation

- Models adapt **on-the-fly** during inference using only target data (entropy minimization, pseudo-labeling).
- Adapting a pretrained model using **only a trained source model**, not the data.
- No access to source data required; useful in privacy-constrained or streaming settings.

3. Multi-source and Multi-target DA

- Leveraging multiple source domains or generalizing to multiple target domains.
- Key challenge: dealing with **conflicting domain shifts** and source reliability.

Trending Topics



4. Out-of-Distribution Generalization

- Blurring lines between DA and OOD robustness.
- Focus on **domain generalization**: training on multiple domains and generalizing to unseen ones without adaptation.

5. Domain Adaptation for Regression / Scientific Applications

- DA research has mostly focused on classification; regression problems (e.g., physics, medical data) are getting more attention.
- Requires new alignment methods and uncertainty-aware losses.

6. Vision-Language Models & Multimodal DA

- Adapting **multimodal representations** (e.g., text+image) to new domains.
- Requires alignment across **both modalities and domains**.



Unsolved Problems



1. Mismatch of Label Distributions

- Most methods assume $P(Y|X)$ remains the same, but in real-world settings it may change → **label shift**, which is under-addressed.

2. Lack of Theoretical Guarantees

- Few practical methods come with **provable generalization bounds** under domain shift.
- Need better **theory for real-world, high-dimensional settings**.

3. Interpretability and Trust

- Black-box adaptations make it hard to understand **what features are being aligned**.
- Especially important in **scientific and safety-critical domains**.



Unsolved Problems



4. Measuring Adaptation Success

- Evaluating whether a model has "adapted well" remains vague.
- Need better **metrics and diagnostics** (e.g., generalization error decomposition, target risk estimation).

5. Continual and Lifelong Adaptation

- Adapting to a **sequence of domains** without forgetting past ones.
- Combines domain adaptation with **continual learning**.

6. Efficient and Scalable Adaptation

- Many DA algorithms are compute-heavy or memory-intensive.
- Need **efficient, lightweight** solutions for deployment in edge devices or real-time systems.

⚙️ Domain Adaptation Problems Quiz !

Scenario 1:

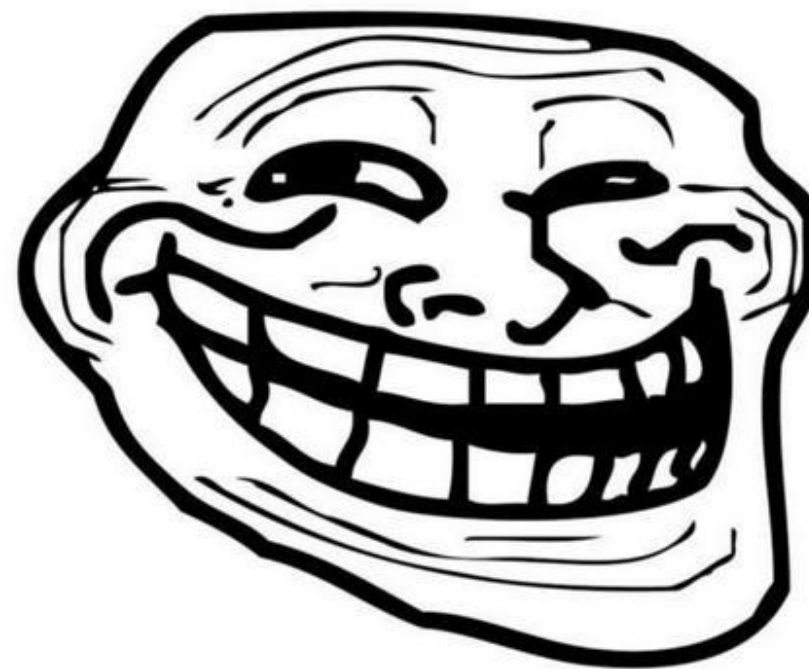
You apply a domain adaptation technique to align galaxy images from two telescopes. After adaptation, your model performs **worse** on the source data and loses its ability to distinguish spiral and elliptical galaxies. What likely went wrong?

Learning rate too small

Overfitting to source data

Too aggressive domain alignment

Answer:



problem?

⚙️ Domain Adaptation Problems Quiz !

Scenario 1:

You apply a domain adaptation technique to align galaxy images from two telescopes. After adaptation, your model performs **worse** on the source data and loses its ability to distinguish spiral and elliptical galaxies. What likely went wrong?

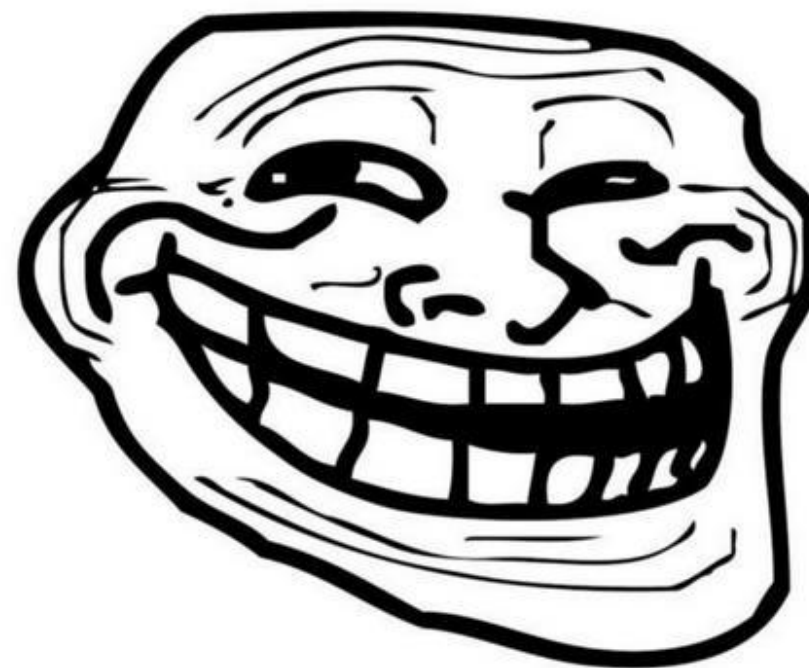
Learning rate too small

Overfitting to source data

Too aggressive domain alignment

Answer:

This is a classic **over-alignment** issue: when features are overly forced to match across domains, the model can discard **discriminative features** that are important for the task. This is especially risky in scientific domains where small differences matter, or domain overlap is small. Careful balance between alignment and task supervision is crucial.



problem?

Domain Adaptation Problems Quiz !

Scenario 2:

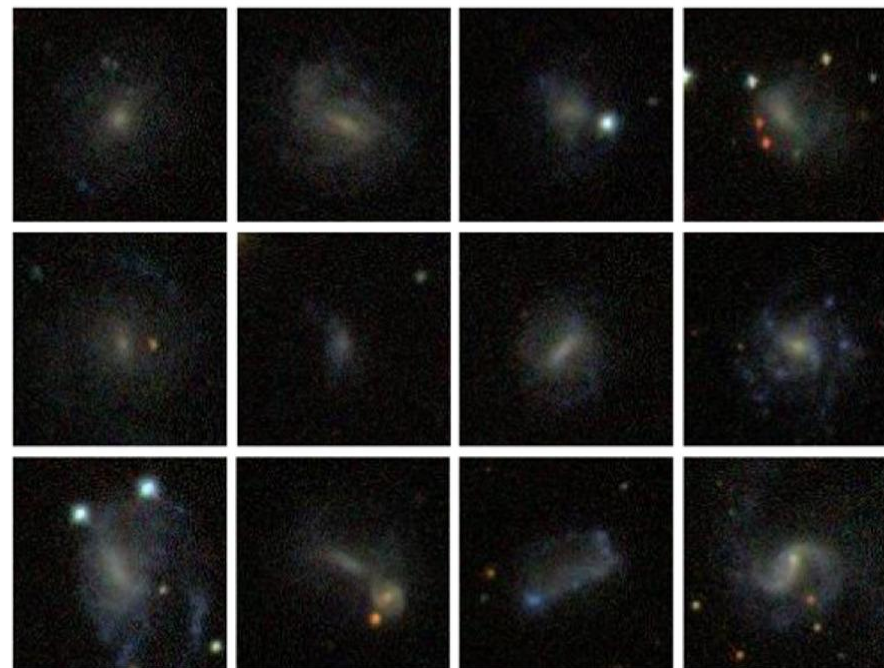
You trained a classifier on old survey data where low-surface brightness galaxies are rare, but in the new LSST survey, they dominate. Your adapted model performs poorly on the target domain, despite good feature alignment. What is the most likely cause?

We didn't account for label distribution shift

Feature extractor wasn't pre-trained

Training didn't converge

Answer:



Domain Adaptation Problems Quiz !

Scenario 2:

You trained a classifier on old survey data where low-surface brightness galaxies are rare, but in the new LSST survey, they dominate. Your adapted model performs poorly on the target domain, despite good feature alignment. What is the most likely cause?

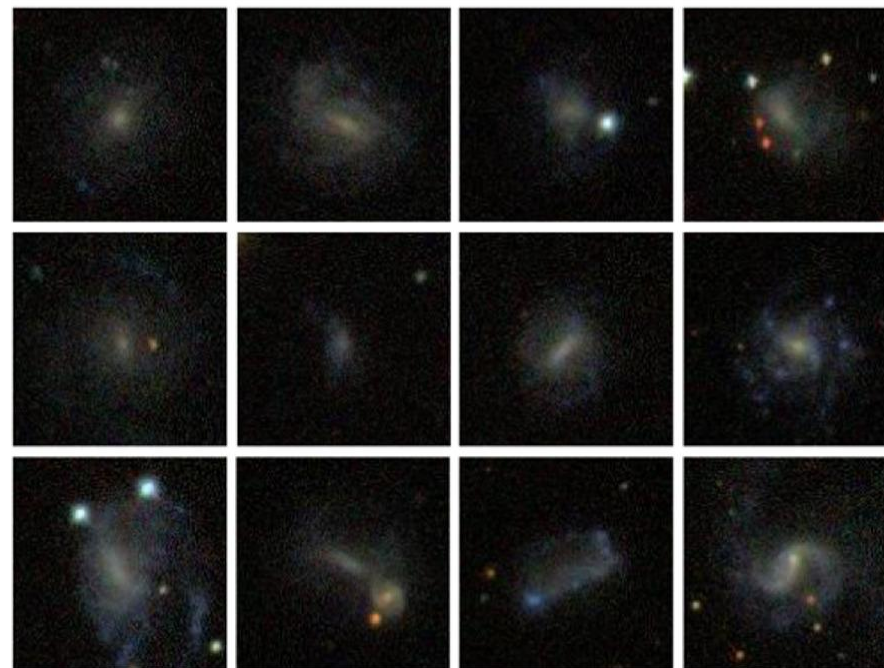
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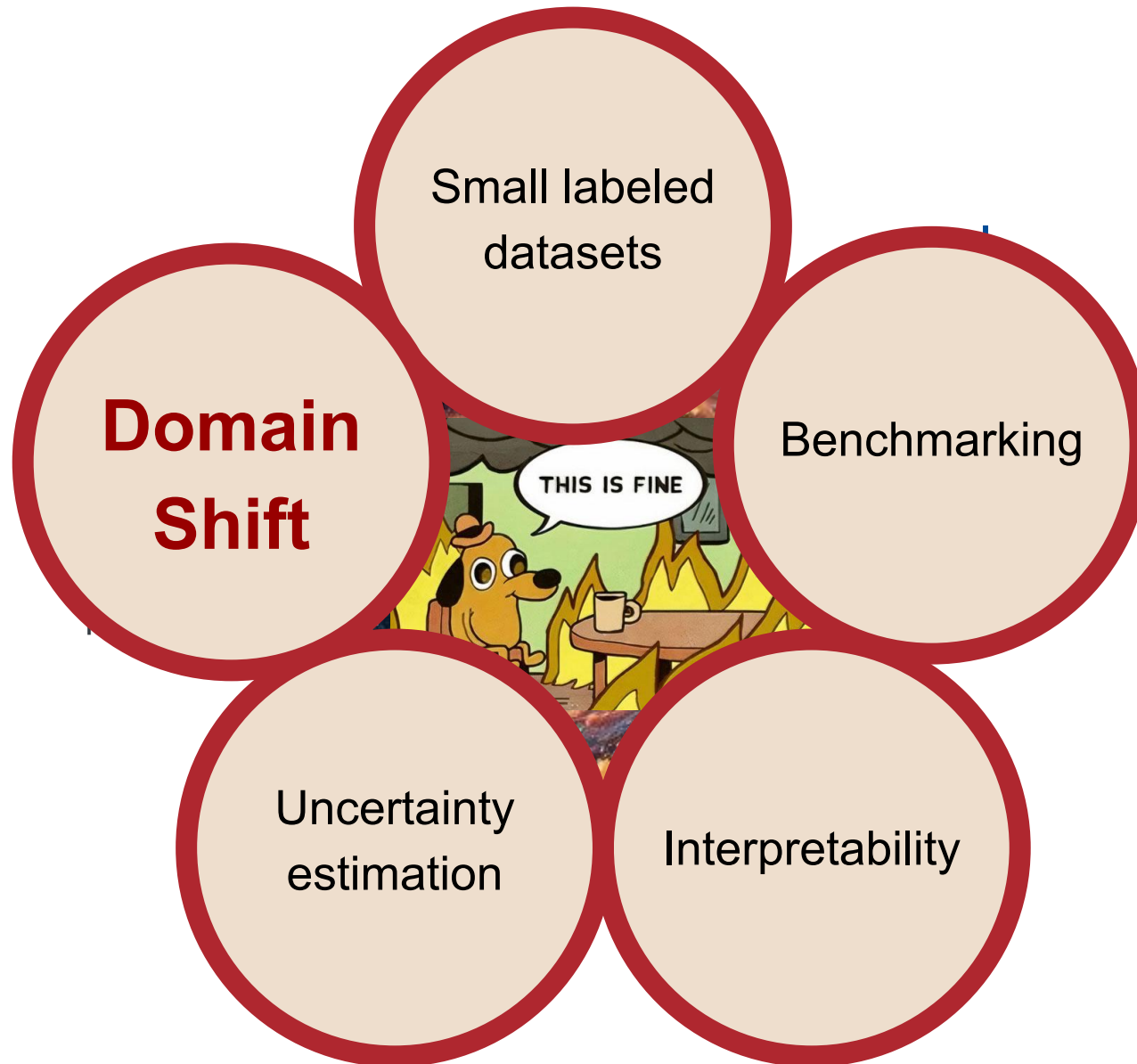
Answer:

Most domain adaptation methods assume that $P(\mathbf{Y}|\mathbf{X})$ stays the same (covariate shift). If instead $P(\mathbf{Y})$ or $P(\mathbf{Y}|\mathbf{X})$ changes, models trained on one distribution may misclassify dominant classes in the new domain. Solutions like target-prior correction, self-supervision or instance reweighting are needed.





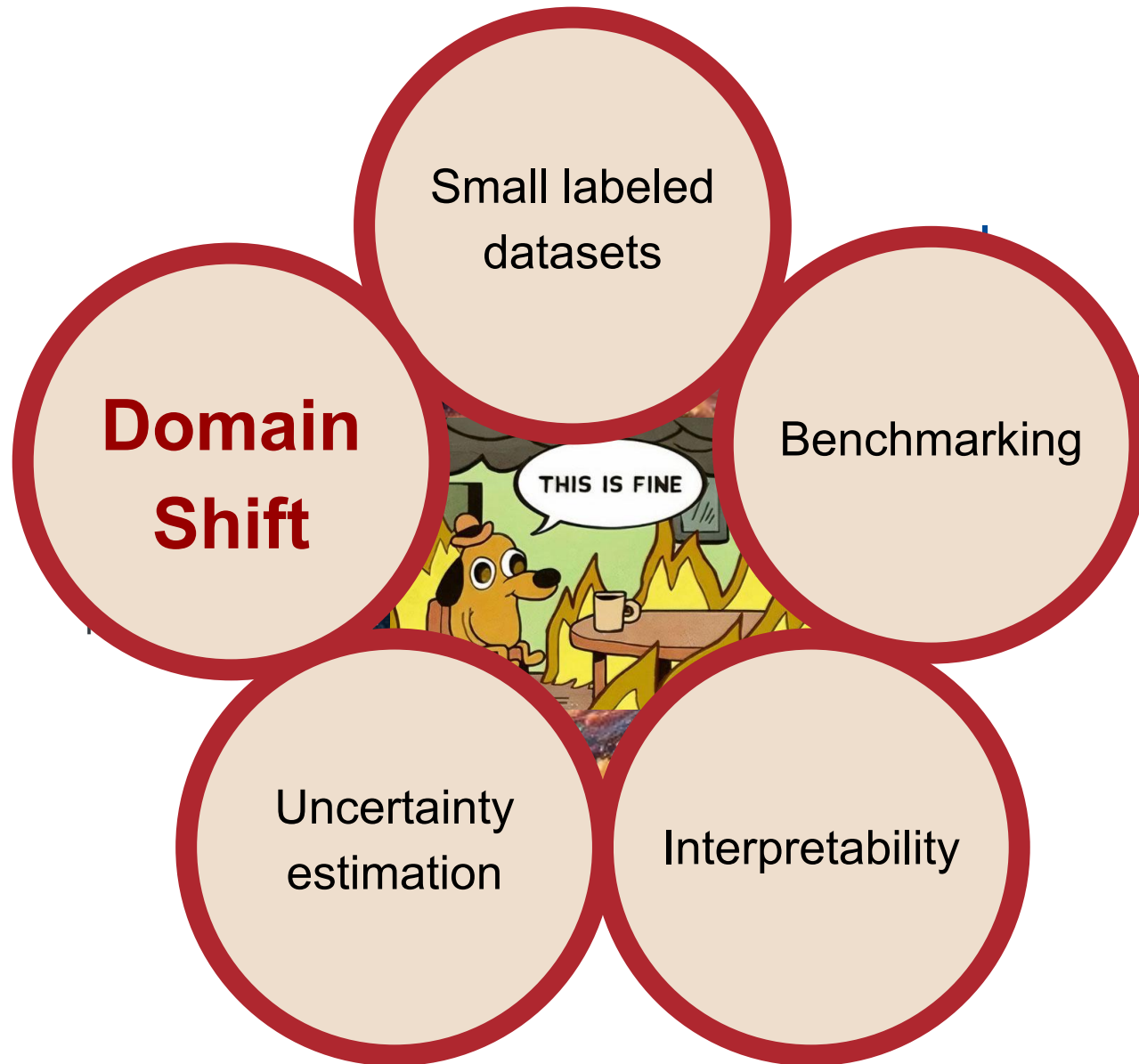
Summary



- Era of big astro surveys - **large amount of data but not all of it labeled** (i.e. suitable for easy AI applications).
- Simulations and old data are different - **domain shift problem!**
- **Domain Adaptation** can help but it cannot solve all problems.
- Hands on session:
 - Learn about different DA methods & how to tune them.



Summary



THANK YOU!

aleksand@fnal.gov

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- **Domain Adaptation** can help but it cannot solve all problems.
- Hands on session:
 - Learn about different DA methods & how to tune them.

LSST DSFP, June 2026

Shift Happens: Building Robust AI Models with Domain Adaptation

TUTORIAL

Aleksandra Ciprijanvoic

Associate Scientist, Fermilab
aleksand@fnal.gov



U.S. DEPARTMENT
of **ENERGY**

Fermi National Accelerator Laboratory is managed by
FermiForward for the U.S. Department of Energy Office of Science

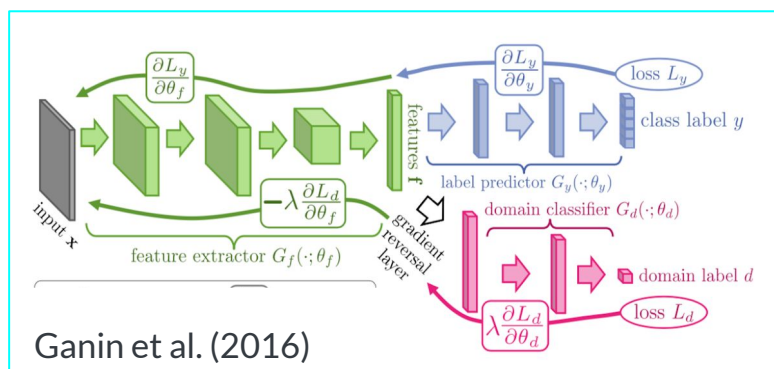


Combining Datasets

DOMAIN ADAPTATION

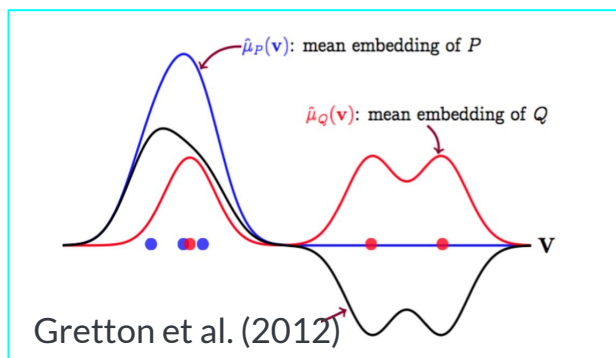
Align data distributions in the latent space of the network by forcing the network to **find more robust domain-invariant features**.

Adversarial



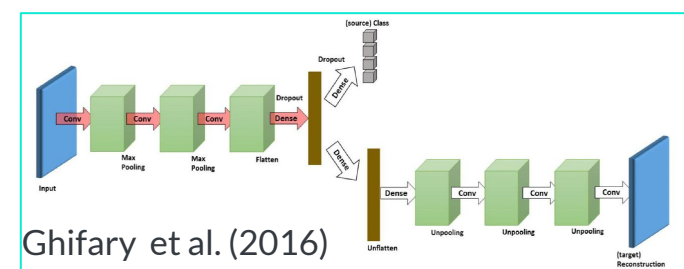
Using domain discriminators to encourage domain confusion through an adversarial objective.

Distance-based methods



Minimize a distance metric between source and target data.

Reconstruction-based



Data reconstruction as an auxiliary task to ensure feature invariance.

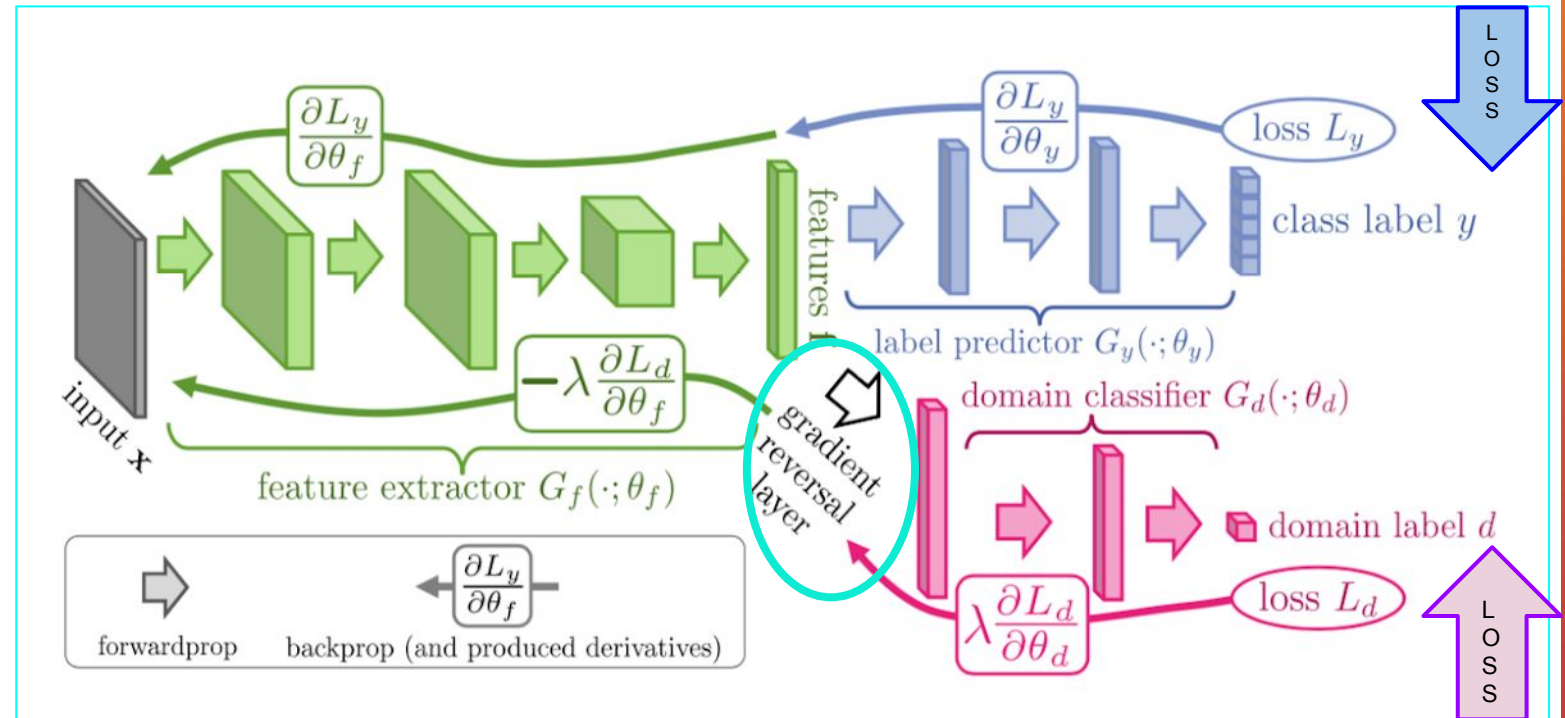


Adversarial Training

Domain Adversarial Neural Networks - DANNs

DANN - feature extractor + label predictor + domain classifier

- **Gradient reversal layer** - multiplies the gradient by a negative constant during the backpropagation.
- Results in the extraction of **domain-invariant features**.
- Only source domain images are labeled during training.



Ganin et al. (2016)

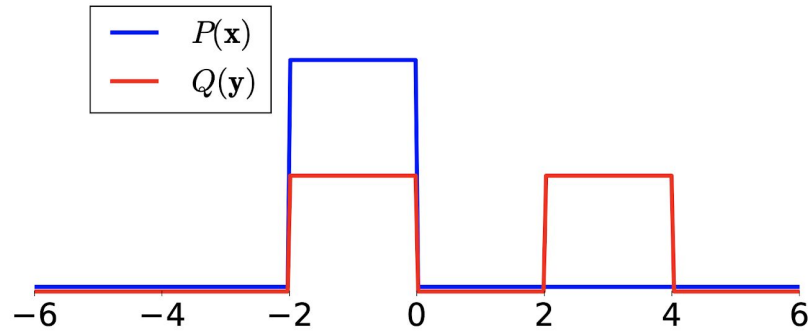


Distance-Based Training

Maximum Mean Discrepancy - MMD

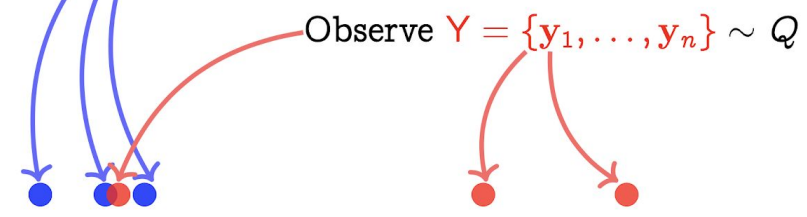
Smola et al. (2007)
Gretton et al. (2012)

Are P and Q different?



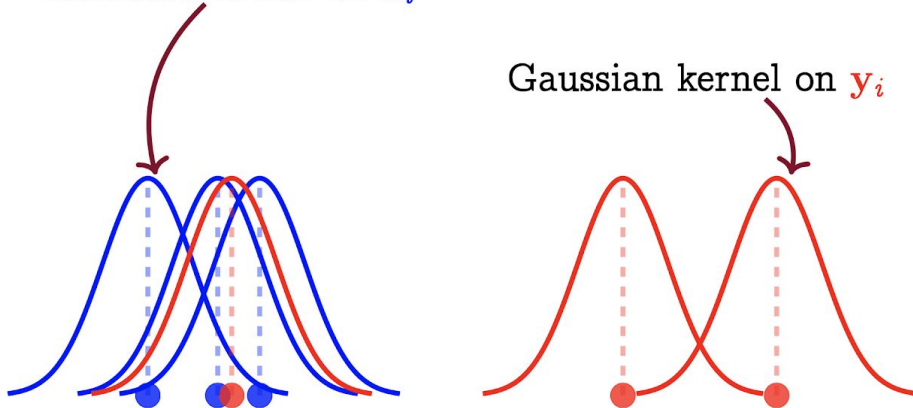
Observe $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \sim P$

Observe $Y = \{\mathbf{y}_1, \dots, \mathbf{y}_n\} \sim Q$



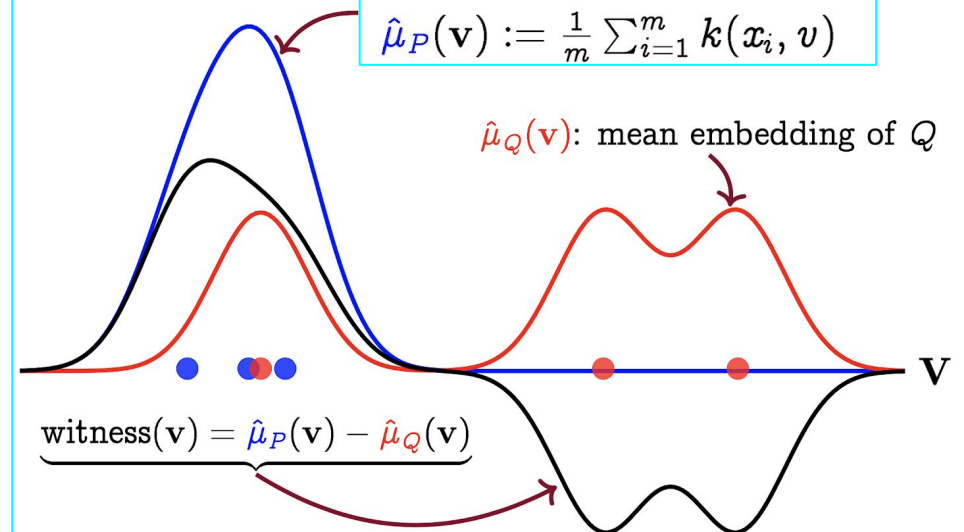
Gaussian kernel on $\mathbf{x}_i$

Gaussian kernel on $\mathbf{y}_i$



$$\hat{\mu}_P(\mathbf{v}) := \frac{1}{m} \sum_{i=1}^m k(\mathbf{x}_i, \mathbf{v})$$

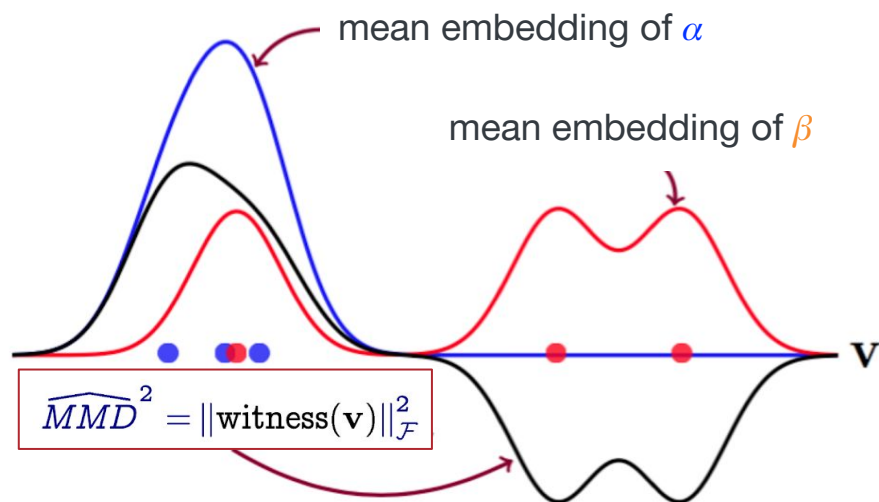
$\hat{\mu}_Q(\mathbf{v})$: mean embedding of Q



Maximum Mean Discrepancy (MMD)

Gaussian (RBF) kernel

$$k(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$



$$\text{MMD}^2(\alpha, \beta) = \mathbb{E}_{x, x' \sim \alpha}[k(x, x')] + \mathbb{E}_{y, y' \sim \beta}[k(y, y')] - 2\mathbb{E}_{x \sim \alpha, y \sim \beta}[k(x, y)]$$

Maximum Mean Discrepancy (MMD)

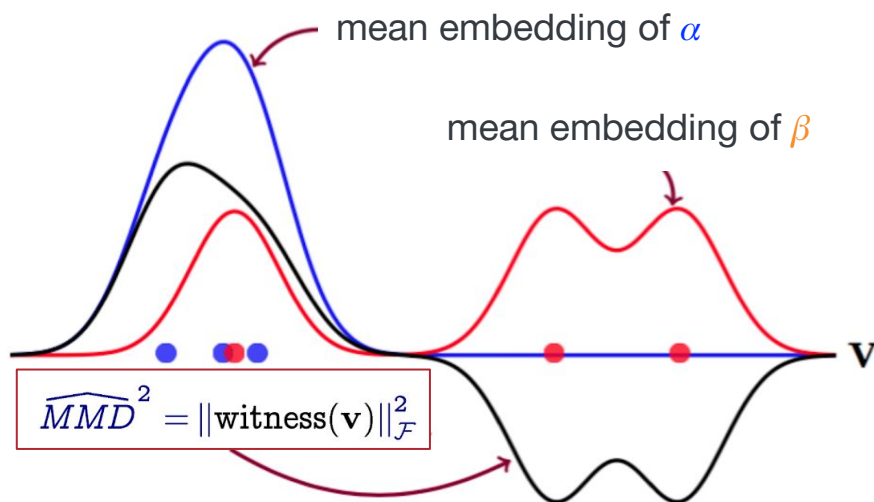
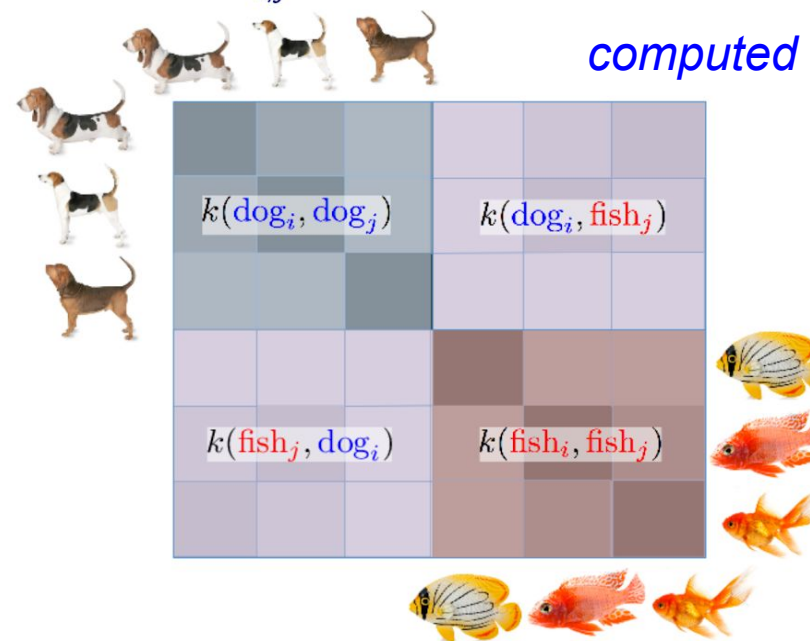
Smola et al. (2007)
Gretton et al. (2012)

Gaussian (RBF) kernel

$$k(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

$$\widehat{MMD}^2 = \frac{1}{n(n-1)} \sum_{i \neq j} k(\text{dog}_i, \text{dog}_j) + \frac{1}{n(n-1)} \sum_{i \neq j} k(\text{fish}_i, \text{fish}_j) - \frac{2}{n^2} \sum_{i,j} k(\text{dog}_i, \text{fish}_j)$$

computed in $O(n^2)$



$$\text{MMD}^2(\alpha, \beta) = \mathbb{E}_{x, x' \sim \alpha} [k(x, x')] + \mathbb{E}_{y, y' \sim \beta} [k(y, y')] - 2\mathbb{E}_{x \sim \alpha, y \sim \beta} [k(x, y)]$$

Maximum Mean Discrepancy (MMD)

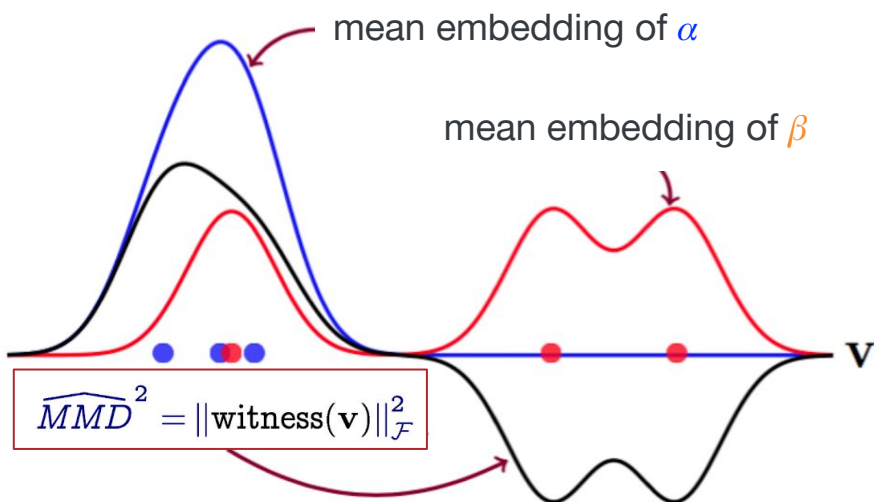
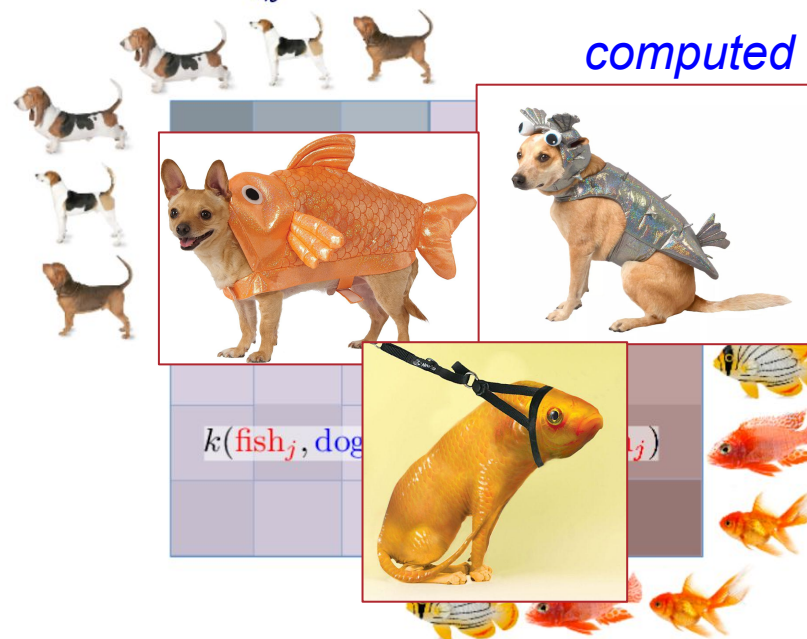
Smola et al. (2007)
Gretton et al. (2012)

Gaussian (RBF) kernel

$$k(x, y) = \exp\left(-\frac{\|x - y\|^2}{2\sigma^2}\right)$$

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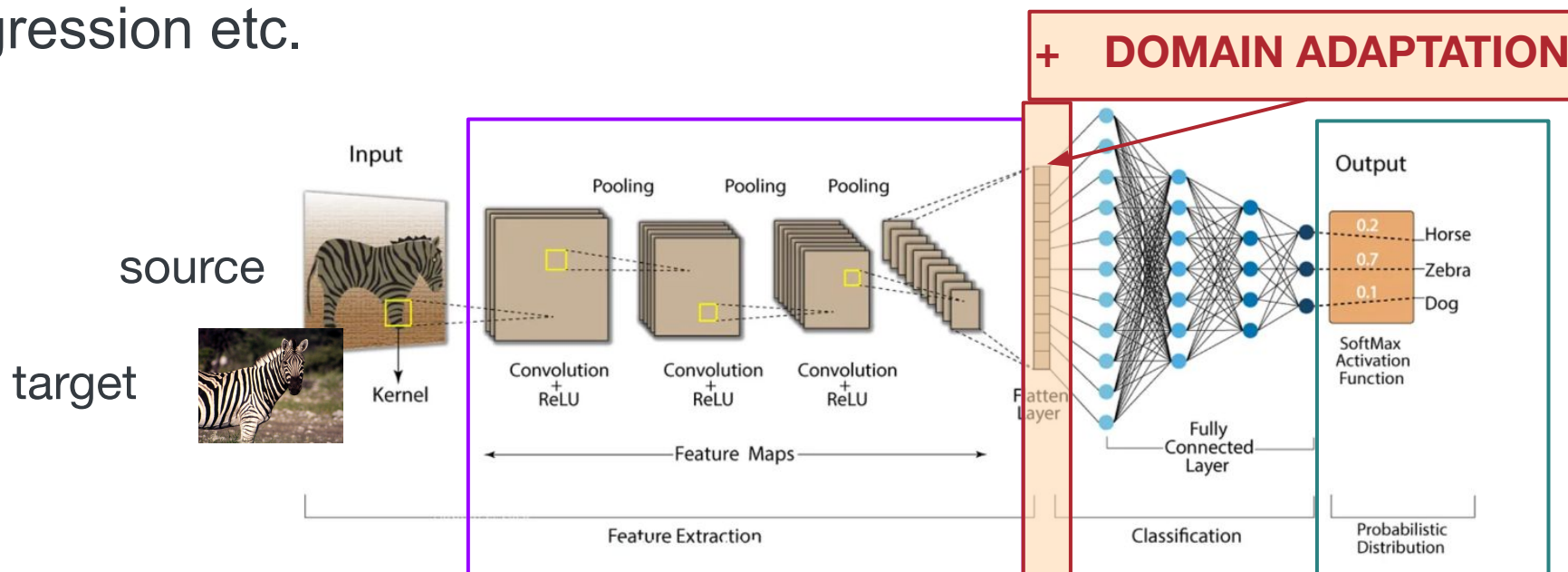
computed in $O(n^2)$



$$MMD^2(\alpha, \beta) = \mathbb{E}_{x, x' \sim \alpha} [k(x, x')] + \mathbb{E}_{y, y' \sim \beta} [k(y, y')] - 2\mathbb{E}_{x \sim \alpha, y \sim \beta} [k(x, y)]$$

✚ Tutorial Intro - Domain Adaptation Training

- **Feature extractor** encodes both source and target data to get their latent embeddings. **Latent embeddings are used for domain alignment.** This leads to the extraction of domain-invariant features by the feature extractor. We will do that by:
 - minimizing a distance metric between source and target embeddings
 - employing **adversarial training**
- **Source domain logits** are used for supervised learning task - classification, regression etc.



Tutorial Intro - Data and Domain Shift

Data: source domain $\mathcal{D}_s = \{x_s^{(i)}, y_s^{(i)}\}_{i=1}^N$ target domain $\mathcal{D}_t = \{x_t^{(i)}\}_{i=1}^N$

Covariate Shift: $x_s^{(i)} \sim p_s(x)$ and $x_t^{(i)} \sim p_t(x)$ but $p(y_s^{(i)} | x_s^{(i)}) = p(y_t^{(i)} | x_t^{(i)})$

Neural Network: $f_\theta = \phi_\theta \circ g_\theta$ for the mapping $\mathbb{R}^{n \times n} \rightarrow \phi_\theta \rightarrow z \rightarrow g_\theta \rightarrow \mathbb{R}^d$

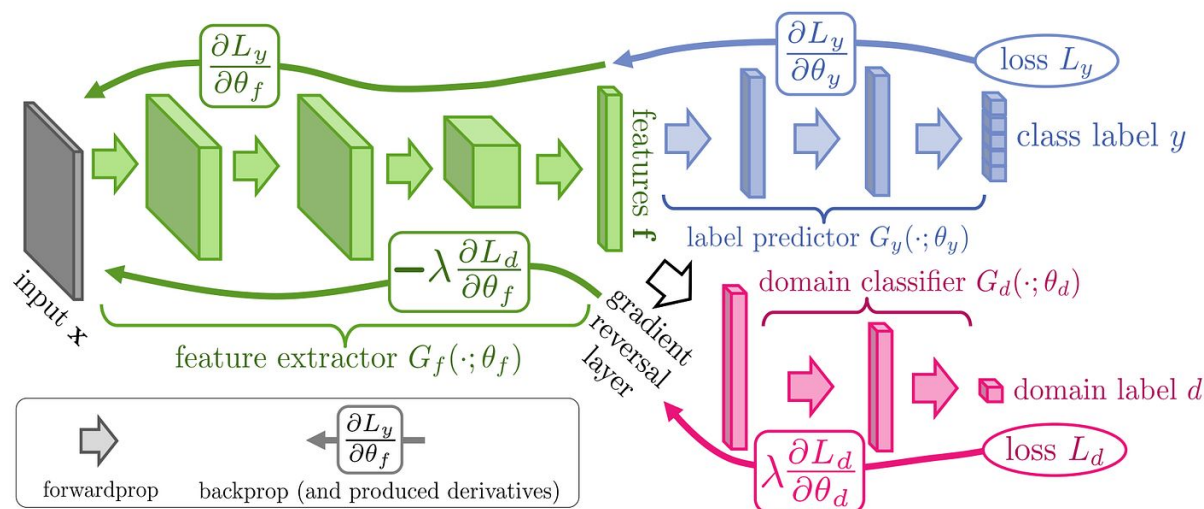
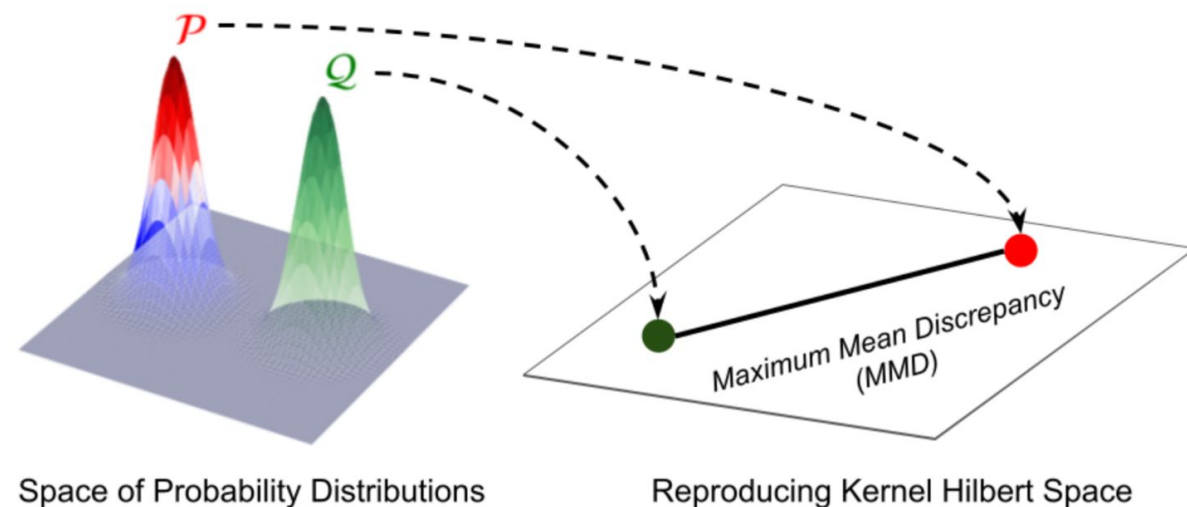
Training: $\mathcal{L}_{\text{tot}} \propto \underbrace{\mathcal{L}_{\text{CE}}(y_s, \hat{y}_s)}_{\text{supervised}} + \underbrace{\mathcal{L}_{\text{DA}}(z, z^*)}_{\text{unsupervised}}$ semi-supervised

Tutorials: $\mathcal{L}_{\text{DA}} \in \{\text{MMD}, \text{Adversarial}\}$

$\mathcal{D}_s, \mathcal{D}_t \in \{\text{MNIST}, \text{Galaxy Images}\}$

Tutorial - Goals

- Goal:** Compare maximum mean discrepancy (MMD) and adversarial methods for DA from MNIST $\rightarrow$ MNIST-M or Galaxy Mergers (pristine $\rightarrow$ noisy) datasets
- Focus:** Conceptual understanding of the DA methods (not PyTorch syntax)
- Pay attention to:** TO DO comments, stretch questions, further exploration ideas
- Chat and work with your peers, work at your own pace! There is plenty to do.





Tutorial 1 - Instructions

- **Location:** <https://github.com/LSSTC-DSFP/Session-25/tree/main/Day4>
- **Pick:** GPU (if you have it on your laptop or in Colab)
 - a) Everything should run smoothly, but galaxy merger data loading might take ~10-15 mins
- **Run:** Most packages are typical, but if any fails, pip install



Now that you have mastered the basics...

you can also consider using or becoming one of our ShiftKit beta testers

<https://alekscipri.github.io/ShiftKit/>

Overview

Source-Only vs MMD vs DANN

MMD vs LMMD comparison

Swapping the model

Custom dataset

Tuning λ

Regression DA

API Reference

Data

Models

Methods

Diagnostics

Citation

Credits



A lightweight, modular **domain adaptation** framework for PyTorch. Transfer knowledge from a labelled *source* domain to an unlabelled *target* domain using deep latent-space alignment.

>

